LM01 Organizational Forms, Corporate Issuer Features, and Ownership

Ver 1.0

1. Introduction

This learning module covers:

- Different organizational forms of business – sole proprietorship, general partnerships, limited partnerships, and corporations.
- Key features of corporate issuers
- Differences between publicly and privately owned corporate issuers

2. Organizational Forms of Businesses

Common forms of business structures include:

- Sole proprietorship
- General partnership
- Limited partnership
- Corporation

To understand the differences between these business structures we will focus on four areas:

- <u>Legal relationship</u>: the legal relationship between the owners and the business.
- <u>Owner-operator relationship</u>: the relationship between the owners of the business and those who operate the business.
- <u>Business liability</u>: the extent to which owners are liable for the actions undertaken by the business.
- <u>Taxation</u>: the tax treatment of profits generated by the business.
- <u>Access to financing</u>: the ability to raise capital to fund expansion and distribute risks.

Sole Proprietorship (Sole Trader)

A sole proprietorship is the most basic type of business structure. In a sole proprietorship, the owner personally funds the capital required to operate the business and retains full control over the business's operations. The owner also fully participates in the financial returns and risks of the business. An example of a sole proprietorship is a family-owned store.

The key features of a sole proprietorship are:

- <u>Legal relationship</u>: It has no legal identity and is considered an extension of the owner.
- <u>Owner-operator relationship</u>: It is an owner operated business and the owner retails full control of the business.
- <u>Business liability</u>: The owner has unlimited liability. He retains all risk associated with the business and can be held financially responsible for all debt the business owes.
- <u>Taxation</u>: Profits from the business are taxed as personal income.

- Access to financing: Sole proprietorships are preferred for small scale business due to their simplicity and flexibility. However, their main drawback is that the business is constrained by the owner's ability to access capital and assume risk.

General Partnership

A general partnership is similar to a sole proprietorship with the important distinction that they have two or more owners called 'partners'. The roles and responsibilities of partners are outlined in a partnership agreement. As compared to sole proprietorships, this structure allows for additional resources to be brought into the business. The business risk is also distributed among a larger group of individuals. Examples of general partnerships include professional services businesses such as law, accounting, and financial advisory firms.

The key features of a general partnership are:

- Legal relationship: It has no legal identity. The partnership agreement defines the ownership of the business.
- Owner-operator relationship: The business is operated by partners. Often the partners bring complementary expertise, such as business development, financial acumen, operations, or legal/compliance, and share responsibility in running the business.
- Business liability: The partners share all risk and business liability. If one partner is unable to pay their share of the business's debts, the remaining partners are fully liable.
- Taxation: Profits from the business are taxed as personal income of the partners.
- Access to financing: Like sole proprietorships, the potential for growth for a general partnership is limited by the partners' ability to source capital, their expertise, as well as their collective risk tolerance.

Limited Partnership

A limited partnership is a special type of partnership that has at least one 'general partner' (GP) with unlimited liability, who is responsible for managing the business. The remaining partners are 'limited partners' (LPs) with limited liability i.e., they can only lose up to the amount invested in the business. All partners share profits, with general partners typically getting a larger share. An example of a limited partnership is a private equity fund.

The key features of a limited partnership are:

- Legal relationship: It has no legal identity. The partnership agreement defines the ownership of the business.
- Owner-operator relationship: The GP operates the business. LPs have no control over the operation of the business. They cannot replace the GP in the event he runs the business poorly or fails to act in the interest of the LPs.
- Business liability: GP has unlimited liability, while LPs have limited liability.
- Taxation: All partners share profits and the profits are taxed as personal income.

- Access to financing: Business growth is limited by the financing capabilities and risk appetite of the partners. Also, GP's competence and integrity in running the business affects the business.

Corporation (Limited Companies)

A corporation is an evolved model of the limited partnership. It is also called a limited liability company (LLC) or limited company in many countries.

Like a limited partnership, owners in a corporation have limited liability; however, corporations provide greater access to capital and expertise needed to fuel growth. Examples of corporations are national or multinational conglomerates.

The two main types of corporations are:

- Public limited companies
- Private limited companies

The main difference between the two are the number of shareholders and whether the shares are listed on a stock exchange. In some countries like the UK, a corporation is categorized as public if the shareholders are greater than 50. While in many other countries, like the US, a corporation is categorized as public if the company shares are listed on an exchange.

We will discuss the key features of a corporation in the next section.

3. Key Features of Corporate Issuers

Legal relationship:

A corporation is a legal entity separate and distinct from its owners. Corporations have many of the same rights and responsibilities as an individual. For example, corporations can enter into contracts, hire employees, borrow and lend money, and pay taxes.

Large corporations frequently conduct business in many different geographic regions and are subject to regulatory jurisdictions where either:

- the company is incorporated,
- business is conducted, or
- the company's securities are listed

Owner-operator separation:

A key feature of corporations is the separation between those who own the business – the shareholders, and those who operate it – the board of directors and company management.

The shareholders elect a board of directors to oversee business operations. The board hires the CEO and senior management for day-to-day operations of the company. This separation of operating control from ownership enables corporations to obtain financing from a large number of investors who are not required to have any expertise in operating the business.

While the board and company management are supposed to act in the best interest of shareholders, conflicts of interest can occur when management places their own interests, or the interests of other stakeholders, above those of the shareholders. To prevent such conflicts and mismanagement of business, corporate governance policies and practices are put in place. If the board or management does not perform as expected, shareholders have the ability to enact change through voting rights attached to their shares. This ability to change things through voting is the key difference between a corporation and a limited partnership.

<u>Business liability</u>:

Owners of a corporation have limited liability. The maximum amount that they can lose is what they invested in the company. Owners also have a residual claim on the company's net assets after its liabilities have been paid. They can thus participate in the growth of the company.

<u>Access to financing</u>:

The corporate form of business structure is preferred when capital requirements are greater than what could be raised through other business structures. Corporations can raise two types of capital:

- Ownership capital (equity)
- Borrowed capital (debt)

Both equity shareholders and debt security holders are referred to as investors in the corporation's securities. However, equity shareholders purchase an ownership stake that entitles them to a residual claim in the corporation. Whereas, bondholders are lenders to the corporation and do not have any ownership entitlement.

<u>Taxation</u>:

In many jurisdictions, corporate profits are taxed twice (double taxation): once at the corporate level and again at the individual level when profits are distributed as dividends to the owners.

Example from the curriculum

The French company Elo (previously known as Auchan Holding) generated operating income of €838 million and paid corporate taxes of €264 million. Investors in France also pay a 30% tax on dividends received. If Elo had distributed all of its after-tax income to investors as a dividend, what would have been the effective tax rate on each euro of operating earnings?

Solution:

Operating Income	€838
Corporate Taxes (31.5%)	€264
After-Tax Income	(€838 – €264) = €574
Distributed Dividend	€574
Investor Dividend Tax (30%)	€574 × 0.3 = €172.2
Effective Tax Rate	(€264 + €172.2) / €838 = 52.1%

If the remaining after-tax income of €574 million was paid to investors as a dividend, investors would pay €172.2 million in taxes on the dividends received. Total taxes paid would be €436.2 million (€264 million at the corporate level plus €172.2 million at the personal level), resulting in an effective tax rate of 52.1%.

Exhibit 6 from the curriculum provides a summary of the key distinctions between organizational forms of business.

Feature	Sole Proprietor	General Partnership	Limited Partnership	Corporation
Legal Identity	No separate legal identity; extension of owner	No separate legal identity; extension of partner(s)	No separate legal identity; extension of partner(s)	Separate legal entity
Owner–Operator Relationship	Owner operated	Partners operated	GP operated	Board and management operated
Owner Liability	Sole unlimited liability	Shared unlimited liability	GP has unlimited liability; LPs have limited liability	Limited liability
Taxation	Pass-through: Profits taxed as personal income	Pass-through: Profits taxed as personal income	Pass-through: Profits taxed as personal income	Corporation income taxed; distributions (dividends) taxed as personal income
Access to Financing	Limited by owner access to capital	Limited by partner access to capital	Limited by GP/LP access to capital	Unbounded access to capital, unlimited business potential

4. Publicly vs. Privately Owned Corporate Issuers

In this section, we will compare public and private companies with respect to:

- Exchange listing, liquidity, and price transparency
- Share issuance
- Registration and disclosure requirements

Exchange Listing, Liquidity, and Price Transparency

Public companies are usually listed on an exchange. This allows ownership to be easily transferred. New investors can become shareholders in a public company by simply executing a buy order. Similarly, exiting investors can liquidate their ownership by executing a sell order. These trades can be executed within a matter of seconds.

Each trade between a buyer and seller can change the share price. We can plot the share price over time to see how the company's value changes. We can also see how the company's value is impacted by significant news about the company specifically or about the general state of the economy.

In contrast, private companies are not listed on an exchange, and, therefore have no observable stock price, which makes their valuation challenging. Transactions between buyers and sellers have to be privately negotiated and transferring ownership is much more difficult. Investments in private companies are usually locked up until the company is acquired by another company, or it goes public.

However, the potential returns in private companies can be much larger than public companies. This is because investors in private companies usually join early in the company's life cycle and they have greater control over management.

Share Issuance

Public companies typically raise very large amounts of capital from many investors through an IPO and through additional issues after listing. These investors then actively trade shares among themselves in the secondary market.

In contrast, private companies raise much smaller amounts from far fewer investors who have much longer holding periods. Because of the higher risks, only accredited investors are permitted to invest in private companies. Accredited investors are those who are sophisticated enough to take greater risks and to have a reduced need for regulatory oversight and protection. To be considered accredited, an investor must have a certain level of income or net worth or possess a certain amount of professional experience or knowledge.

Registration and Disclosure Requirements

Public companies are subject to greater regulatory and disclosure requirements. They are required to disclose their financial reports as well as other information that may affect stock price, such as - major changes in the holding of voting rights, any stock transactions made by officers and directors etc. These documents are made available to the general public, not just to the company's investors. The primary purpose of this type of disclosure is to make it easier for investors and analysts to assess the company.

In contrast, private companies are generally not required to make such disclosures and they are not subject to the same level of regulatory oversight.

Going from Private to Public

A private company can go public in the following ways -

- IPO: To complete an IPO, a company must meet the specific listing requirements of the exchange. The IPO is facilitated by investment banks who underwrite (guarantee) the offering.
- Direct listing (DL): A DL differs from an IPO in two key ways – (1) no underwriter is involved and (2) no new capital is raised. Instead, the company is simply listed on an exchange and shares are sold by existing shareholders. As compared to an IPO, a DL is much faster to execute and involves lower costs.
- Acquisition: Companies can also go public through acquisitions. This may occur indirectly when a private company is acquired by a larger public company.

  Another means of acquisition is through a special purpose acquisition company (SPAC). A SPAC is a shell company, often called a "blank check" company, because it exists solely for the purpose of acquiring an unspecified private company sometime in the future.

  SPACs raise capital through IPOs and deposit the proceeds in a trust account. They have a finite time (e.g., 18 months) to complete a deal; otherwise, the proceeds are returned back to the investors.

Going from Public to Private

Some public companies may also choose to go private. To accomplish this an investor has to acquire all of the company's shares and delist it from the exchange. Investors must typically pay a premium above the current share price and often use debt to finance the acquisition. "Go-private" transactions are initiated when investors believe that the public market is undervaluing the shares and when financing costs are sufficiently low to make such transactions attractive.

The number of public companies is increasing in many emerging economies, while it is decreasing in many developed economies.

Emerging economies usually have higher growth rates and are transitioning from closed to open market structures. Therefore, the number of public companies are increasing as these economies grow larger.

In developed economies, the number of public companies is trending downwards because:

- M&A activities: When a public company is acquired by another private or public company, one less public company exists.
- Because of the thriving venture capital, private equity, and private debt markets, it has become easier for private companies to access the capital they need without having to

go public. This helps companies avoid regulatory burden and the associated compliance costs of going public.

- Many private companies simply choose to remain private because it allows the owners and management to maintain control over the company.

Summary

LO: Compare the organizational forms of businesses.

Feature	Sole Proprietor	General Partnership	Limited Partnership	Corporation
Legal Identity	No separate legal identity; extension of owner	No separate legal identity; extension of partner(s)	No separate legal identity; extension of partner(s)	Separate legal entity
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Access to Financing	Limited by owner access to capital	Limited by partner access to capital	Limited by GP/LP access to capital	Unbounded access to capital, unlimited business potential

LO: Describe key features of corporate issuers.

- Legal relationship: It is a legal entity separate and distinct from its owners
- Owner-operator relationship: There is separation between the owners and the operators. The shareholders elect a board of directors to oversee business operations. The board hires the CEO and senior management for day-to-day operations of the company.
- Access to capital: Corporations can raise large amounts of capital through debt or equity.
- Business liability: Owners have limited liability
- Taxation: Corporate profits are taxed twice once at the corporate level and again at the individual level when profits are distributed as dividends.

LO: Compare publicly and privately owned corporate issuers.

Public companies are listed on an exchange which allows ownership to be easily transferred. In contrast, private companies are not listed on an exchange. Transactions between buyers and sellers are privately negotiated which makes ownership difficult to transfer.

Public companies typically raise very large amounts of capital from many investors through an IPO and through additional issues after listing. These investors then actively trade shares among themselves in the secondary market. In contrast, private companies raise much smaller amounts from far fewer investors who have much longer holding periods.

Public companies are subject to greater regulatory and disclosure requirements. They are required to disclose their financial reports as well as other information that may affect stock price, such as - major changes in the holding of voting rights, any stock transactions made by officers and directors etc. In contrast, private companies are generally not required to make such disclosures and they are not subject to the same level of regulatory oversight.

LM02 Investors and Other Stakeholders

Ver 1.0

1. Introduction

This learning module covers:

- Securities issued by corporations – debt and equity, the differences in their risk-return profiles, and the conflicts of interest between debt and equity.
- The various stakeholders of a company and their interests.
- ESG considerations in investment analysis

2. Financial Claims of Lenders and Shareholders

Debt Versus Equity

The key differences between debt and equity financing are:

- Debt represents a contractual obligation on the part of the issuing company. The company is obligated to make the promised interest and principal payments to debtholders. Equity does not involve a contractual obligation.
- Debtholders have a prior legal claim on the company's cash flows and assets. Their claims have to be fully paid before the company can make distributions to equity owners. Equity holders have a residual claim on the company's net asset after all other stakeholders have been paid.
- Interest payments on debt are typically tax-deductible. Whereas, dividend payments on equity are not tax deductible.
- Equity represents a more permanent source of capital. It has no finite term and includes voting rights. In contrast, debt represents a cheaper source of capital. It has a stated finite term and no voting rights.

Debt Versus Equity: Risk and Return

Investor's perspective:

From an investor's perspective, investing in equity is riskier than investing in debt. Debtholders receive predictable coupon payments, whereas payments to shareholders are at the company's discretion.

However, equityholders do have a residual claim, which means they are entitled to whatever firm value remains after other stakeholders have been paid off, giving them unlimited upside potential. Therefore, equityholders have an interest in the ongoing maximization of company value (net assets less liabilities), which directly corresponds to the value of their shareholder wealth.

On the other hand, no matter how profitable a company becomes, debtholders will never receive more than their interest and principal repayment. Their maximum return is capped. They are therefore interested in assessing the likelihood of timely debt repayment and the risk associated with the company's ability to meet its debt obligations.

For both equityholders and debtholders, their initial investment represents their maximum possible loss.

These points are summarized in the table below:

Investor Perspective	Equity	Debt
Tenor	Indefinite	Term (e.g. 10 years)
Return potential	Unlimited	Capped
Maximum loss	Initial investment	Initial investment
Investment risk	Higher	Lower
Desired outcome	Maximize firm value	Timely repayment

Issuer Perspective:

From the company's perspective, issuing debt is riskier than issuing equity. If a company fails to meet its contractual obligations, it may be forced to declare bankruptcy and liquidate.

Although riskier, the cost of debt is lower than the cost of equity (this is because the returns to debtholders are capped). A company generating stable, predictable cash flows generally prefers to borrow money rather than issuing additional equity to raise capital. Issuing more equity also dilutes the upside return for existing equity owners given that residual value must be shared across more owners. However, early-stage companies or companies with unpredictable cash flows that find it difficult to borrow may prefer to raise capital through equity to avoid the risk of default.

In the event of a default, a company does have some options to try and avoid bankruptcy. It can renegotiate more favorable terms with the debtholders. However, if things don't improve, eventually the assets may have to be liquidated to raise as much money as possible to return to the bondholders. Alternatively, the company may be reorganized, with existing shareholders getting wiped out and bondholders becoming the new shareholders of the reorganized company.

These points are summarized in the table below:

Issuer Perspective	Equity	Debt
Capital cost	Higher	Lower
Attractiveness	Creates dilution, may be only option when issuer cash flows are absent or unpredictable	Preferred when issuer cash flows are predictable
Investment risk	Lower, holders cannot force liquidation	Higher, adds leverage risk

Conflicts of Interest among Lenders and Shareholders

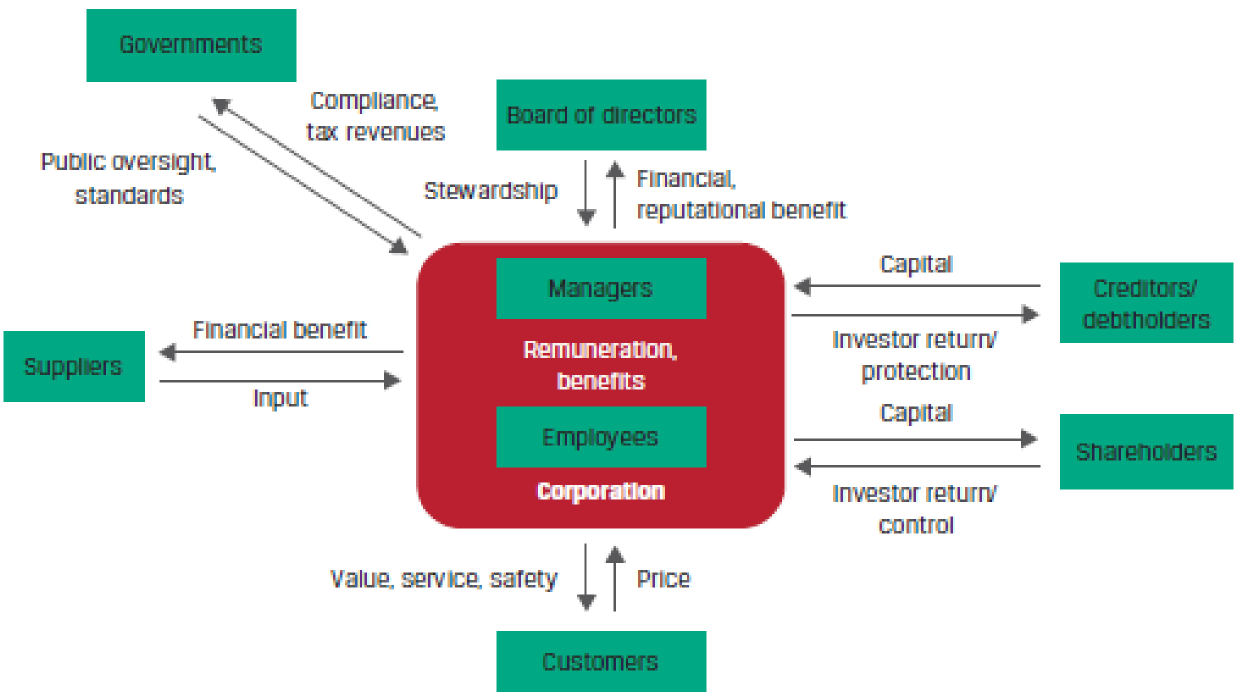
Potential conflicts of interest can occur between debtholders and equityholders. Debtholders would prefer that the company invests in less risky projects that generate predictable cash flows, even if those cash flows are relatively small. Equityholders, on the other hand, would prefer that the company invests in riskier projects with a much higher return potential.

3. Corporate Stakeholders and Governance

A stakeholder is any individual or group that has a vested interest in a company. The primary stakeholder groups of a company include:

- Shareholders and creditors: provide capital to finance the company's activities
- Board of directors: Serves as the steward of the company
- Managers: Execute the strategy set by the board and run day-to-day operations
- Employees: Provide human capital for the day-to-day operations of the company
- Customers: Buy the company's products and services
- Suppliers: Provide raw materials as well as goods and services that cannot be produced efficiently internally.
- Government and regulators: Dictate the rules and regulations governing the company.

Exhibit 6 from the curriculum illustrates these relationships.



Shareholders versus Stakeholders

Traditional corporate governance frameworks were based on the '*shareholder theory*', however, many companies are now moving to corporate governance frameworks based on the '*stakeholder theory*'.

Shareholder theory: The goal of a company is to maximize shareholder returns. The interests of other stakeholders are considered only to the extent that they affect shareholder value.

Stakeholder theory: The company should consider the interests of all stakeholders, not just shareholders. This theory also gives more importance to ESG considerations by making them an explicit objective for the board of directors and management.

We will now look at each stakeholder group in detail.

Shareholders

Shareholders own shares in a corporation and are entitled to certain rights, such as the right to receive dividends and to vote on certain corporate issues.

A shareholder's interests are typically focused on growth in corporate profitability that maximizes the value of the company.

Creditors/Debtholders

Banks and Private Lenders

- They generally hold a company's debt till maturity.
- They typically have direct access to company management and insider information which reduces information asymmetries.
- As compared to public debt holders, banks and private lenders can exert a significant degree of influence on a company.
- Most banks and private lenders prefer companies to have less financial leverage as it implies less risk. They may, however, take very different approaches to lending. Some focus primarily on the collateral supporting their loan; others hold both debt and equity positions in the same company; some have an equity-like focus on the company; and some lend to businesses that they would be interested in owning if the company defaulted.

Public Debtholders (Bondholders)

- They rely on public information and the ratings provided by credit rating agencies to make their investment decisions.
- In return for the capital provided, they expect to receive periodic interest payments and repayment of principal at the end of the term.
- Unlike shareholders, they do not have voting rights.
- They have limited influence over a company's operations.
- They generally seek to minimize downside risk and prefer stability in company's operations and performance. This is in contrast to the interests of shareholders, who are willing to take higher risks for higher potential returns.

Board of Directors

A company's board of directors is elected by shareholders to protect shareholders' interests, monitor the company's operations, and provide strategic direction. The board is also responsible for hiring the CEO and overseeing management performance. A board can be composed on both inside and independent directors.

- Inside directors are major shareholders, founders, and senior managers.
- Independent directors do not have a material relationship with the company in terms of employment, ownership, or remuneration and are chosen for their experience of managing or directing other companies.

There is no standard structure or composition for the board of a company. The number of directors can vary depending on the size, structure, and complexity of operations of the company. However, most corporate governance standards require that board members have a diverse range of expertise, backgrounds, and competencies, with at least one-third of the board being independent.

Staggered board is a commonly followed practice, in which directors are divided into three groups and elected into office in consecutive years, so that all directors are not replaced at the same time. The advantage of a staggered board is that it allows for continuous strategy implementation. The disadvantage is that it limits shareholders' ability to effect an immediate major change in control of the company.

Managers

Managers, led by the CEO are responsible for executing the strategy set by the board. They are also in charge of the company's day to day operations.

Senior managers are typically compensated with a base salary, a short-term cash bonus, and a multi-year incentive plan that includes one or more forms of equity (such as options, time-vested shares, and performance-vested shares). As a result, in addition to acting to protect their employment status, managers may be motivated to maximize the value of their total remuneration.

Employees

Lower-level employees seek fair salary, good working conditions, job security, career development opportunities, promotion, etc.

As compared to managers, equity ownership is a minor part of their compensation. Therefore, they are more interested in the company's long-term stability, survival and growth.

Customers

Customers expect to receive products and services of good quality for the price paid. They also expect after-sales service, support, and guarantee/warranty for the period promised. In return, companies strive to keep their customers happy as this has a direct effect on its revenues. Of all the stakeholders, customers are least concerned about a company's performance.

Suppliers

The primary interest of a supplier is to be paid on time for the products and services delivered to the company. Some suppliers are keen to maintain a good long-term relationship with companies as it is recurring business. Like customers, suppliers typically have an interest in the company's long-term stability.

Governments

Government is a stakeholder as it collects taxes from companies and their employees. Governments seek to protect the interests of the general public and ensure the well-being of their nations' economies.

4. Corporate ESG Considerations

ESG is an acronym for environmental, social and governance issues. The importance of good corporate governance has long been understood by analysts and shareholders. Therefore, the practice of considering governance factors in investment analysis has evolved considerably. However, the practice of considering environmental and social factors in investment analysis has evolved more slowly.

The three main catalysts for growth in ESG investing are:

1. ESG issues are having more financial impacts. Many investors incurred significant losses when the companies they invested in mismanaged ESG issues.
2. A greater number of younger investors are increasingly demanding that their inherited wealth or pension contributions be managed responsibly.
3. Governments have started prioritizing climate change and social policies. New regulations are forcing companies to adapt their business practices to meet more stringent ESG criteria.

Historically, environmental and social issues, such as climate change, air pollution, and societal impacts of a company's products and services, have been treated as negative externalities. However, increased stakeholder awareness and strengthening regulations have led to inclusion of environmental and societal costs in the company's income statement by responsible investors.

Some examples of ESG factors are presented in Exhibit 7 from the curriculum.

Environmental Issues	Social Issues	Governance Issues
• Climate change • Air and water pollution • Biodiversity • Deforestation • Energy efficiency • Waste management • Water scarcity	• Customer satisfaction • Data protection and privacy • Gender and diversity • Employee engagement • Community relations • Human rights • Labor standards	• Board composition • Audit committee structure • Bribery and corruption • Executive compensation • Lobbying • Political contributions • Whistleblower schemes

Typically, a smaller set of factors are material for each company depending on the business segments and geography it operates in.

Governance factors

Corporate governance factors to consider include:

- Company ownership and voting structure
- Relevance of board skills and experience
- Alignment of management compensation with company performance
- Strength of company shareholder rights versus peers
- Company effectiveness in managing long-term risks and sustainability

Corporate governance considerations are reasonably consistent across most companies and industries. However, environmental and social considerations differ greatly across industries. For example, environmental and social concerns for the energy sector will be very different from the concerns for the banking sector.

While evaluating ESG factors, analysts first need to evaluate if a factor is material. An ESG factor is considered to be material when that factor is believed to have an impact on a company's long-term business model.

Some environmental and social factors considered in investment analysis are listed below:

Environmental factors

- Natural resource management.
- Pollution prevention.
- Water conservation.
- Energy efficiency and reduced emissions.
- Existence of carbon assets.
- Compliance with environmental and safety standards.

Social factors

- Human rights and welfare concerns in the workplace
- Data privacy and security
- Access to affordable health care products
- Community impact

Stranded assets: A specific concern among investors of energy companies is the existence of "stranded assets" — carbon-intensive assets that are at risk of no longer being economically viable because of changes in regulation or investor sentiment.

Evaluating ESG-Related Risks and Opportunities

The process of identifying and evaluating ESG-related factors is similar for both equity and debt analyses.

- Once identified, the material impact of an ESG factor must be quantified, i.e. how will the company's future cash flows be impacted?
- If a significant long-term adverse ESG related event is discovered, equity is immediately and disproportionately affected. The company may experience a sharp decline in share prices.
- While debt is also impacted, it is usually less affected than equity unless the company's ability to make interest and principal payments is compromised.
- The effects on debt also vary depending on maturity. For example, some factors may have an impact in the long term but have no impact in the short term. Such factors are not relevant while evaluating short term bonds.

Summary

LO: Compare the financial claims and motivations of lenders and shareholders.

Debt represents a contractual obligation on the part of the issuing company. The company is obligated to make the promised interest and principal payments to debtholders. Equity does not involve a contractual obligation.

Debtholders have a prior legal claim on the company's cash flows and assets. Their claims have to be fully paid before the company can make distributions to equity owners. Equity holders have a residual claim on the company's net asset after all other stakeholders have been paid. This gives them unlimited upside potential.

From an investor's perspective, investing in equity is riskier than investing in debt.

From the company's perspective, issuing debt is riskier than issuing equity. If a company fails to meet its contractual obligations, it may be forced to declare bankruptcy and liquidate.

Potential conflicts of interest can occur between debtholders and equityholders. Debtholders would prefer that the company invests in less risky projects that generate predictable cash flows, even if those cash flows are relatively small. Equityholders, on the other hand, would prefer that the company invests in riskier projects with a much higher return potential.

LO: Describe a company's stakeholder groups and compare their interests.

The primary stakeholders of a company include:

- Shareholders
- Creditors
- Managers and employees
- Board of Directors
- Customers
- Suppliers
- Government/Regulators

LO: Describe environmental, social, and governance factors of corporate issuers considered by investors.

While evaluating ESG factors, analysts first need to evaluate if an information is material. Materiality typically refers to ESG related issues that can affect a company's operation, its financial performance, and the valuation of its securities.

Analysts also consider their investment horizon when deciding which ESG factors to consider in their analysis. Some factors may affect a company's performance in the short term, whereas other issues may be more long term in nature.

Some examples of ESG factors are presented in Exhibit 7 from the curriculum.

Environmental Issues	Social Issues	Governance Issues
• Climate change • Air and water pollution • Biodiversity • Deforestation • Energy efficiency • Waste management • Water scarcity	• Customer satisfaction • Data protection and privacy • Gender and diversity • Employee engagement • Community relations • Human rights • Labor standards	• Board composition • Audit committee structure • Bribery and corruption • Executive compensation • Lobbying • Political contributions • Whistleblower schemes

LM03 Corporate Governance - Conflicts, Mechanisms, Risks, and Benefits

Ver 1.0

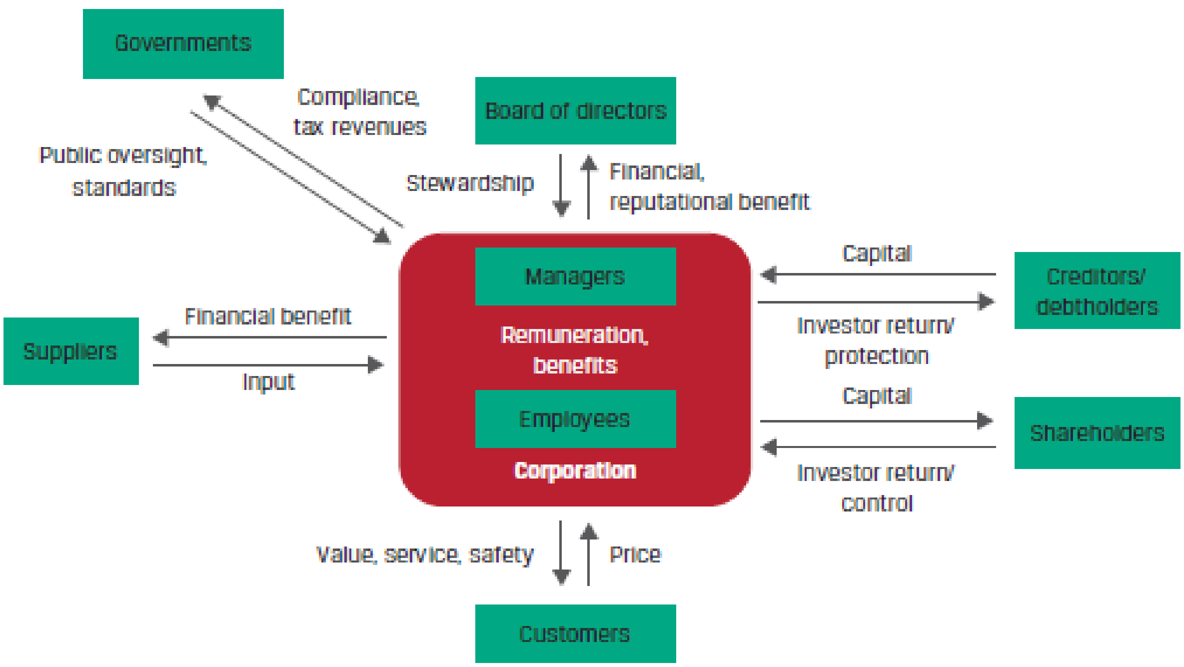
1. Introduction

This learning module covers:

- The principal-agent and other relationships between stakeholder groups
- Corporate governance mechanisms to manage stakeholder relationships
- The potential risks of poor corporate governance and the benefits from effective corporate governance

2. Stakeholder Conflicts and Management

A corporation has a complex ecosystem of stakeholders as illustrated in Exhibit 1 from the curriculum.



A principal-agent relationship arises when a principal hires an agent to carry out a task or a service (e.g. shareholders and managers/directors). An agent is obliged to act in the best interests of the principal and should not have a conflict of interest in performing a task. However, in reality, there are several conflicts of interest that arise in a principal-agent relationship, for example: information asymmetry.

Information asymmetry: Managers have greater access to information about a company's performance and prospects as compared to outsiders such as shareholders and creditors. This information asymmetry reduces shareholders' ability to assess managers' true performance and vote out poor performers. Investors demand higher risk premiums and returns from companies that have greater information asymmetry.

In this section we will discuss a few examples of conflicts between principal-agents and other relationships.

Shareholder and Manager/Director Relationships

In this relationship, shareholders are the principals and managers/directors are the agents. Compensation is the main tool used to align the interest of managers/directors with those of the shareholders. In theory, stock grants/options offered as part of management compensation are intended to motivate managers to maximize shareholder value. However, in practice, the alignment of interests is rarely perfect. Common examples of conflict of interests are:

- **Insufficient effort**: Managers may devote insufficient time to their duties. They may conduct too little monitoring of employees or assert too little control.
- **Entrenchment**: If the overall level of manager/director compensation is excessive, they may try to avoid risk so as to not jeopardize the compensation they have been receiving.
- **Empire building**: If manager/director compensation is tied to the size of the company, it can lead to 'growth for growth's sake'; managers may pursue acquisitions and expansions that do not increase shareholder value.
- **Excessive risk taking**: Because option holders only participate in stock up moves, a compensation package that relies too heavily on stock grants/options may encourage excessive risk taking by management. Similarly, a compensation package with too little or no stock grants/options can have the opposite effect.
- **Self-dealing**: Managers may misuse firm resources to maximize personal benefits (e.g., private planes, club memberships, and personal security), or they may defraud investors by misappropriating assets.

Agency costs: Agency costs arise due to conflicts of interest when an agent makes decisions for a principal. All public companies and large private companies are usually managed by non-owners. Therefore, an agency cost in the context of a corporation is a consequence of a conflict of interest between managers and owners. Since outside shareholders are aware of this conflict, they will take steps that incur costs such as:

- Costs borne by owners to monitor the management of the company, e.g., expenses of the annual report, board of director expenses, etc.
- Costs borne by management to assure owners that they are maximizing the company value, e.g., the implicit cost of noncompete employment contracts and the explicit cost of insurance to guarantee performance.

The better a company is governed, the lower the agency costs.

Controlling and Minority Shareholder Relationships

Corporate ownership structures can be classified as:

- Dispersed: Many shareholders exist and none of them have the ability to individually exercise control over the company.
- Concentrated: An individual shareholder or a group has the ability to exercise control over the company (known as controlling shareholders)

Conflicts of interest may arise between controlling shareholders and minority shareholders. For example:

- Controlling shareholders may have a large percentage of their wealth tied up in company stock and would want management to diversify the company to achieve stability. Minority shareholders with diverse portfolios, on the other hand, would prefer management to focus on the core business because they can diversify cheaply on their own.
- Controlling shareholders may have a multi-decade perspective, whereas some minority shareholders may seek quick gains from cost cutting, selling assets, or share repurchases.

Share ownership alone may not necessarily reflect whether the control of a company is dispersed or concentrated. Controlling shareholders may be either majority shareholders or minority shareholders.

- Majority shareholders: Own more than 50% of a company's shares.
- Minority shareholders: Own less than 50% of a company's shares.

Dual-Share Classes

An equity structure with multiple share classes tends to assign superior voting powers to one class and limited voting rights to other classes. The multiple-class structure allows controlling shareholders avoid dilution of their voting power when new shares are issued and to retain control of board elections, strategic decisions, and all other significant voting matters for an extended period of time — even if their ownership level falls below 50%.

Shareholder vs. Creditor (Debtholder) Interests

There is a conflict of interest between the two suppliers of capital to a company under the following circumstances:

- Distribution of dividends: Creditors are concerned if a company pays excess dividends to shareholders that may impair its ability to service debt.
- Risk tolerance: Shareholders have a higher risk tolerance and prefer a company takes on more risk to generate higher returns. The better the performance of a company, the higher is the return shareholders can expect. Creditors are conservative and prefer a stable operating cash flow over higher returns as they do not have a claim to residual income.
- Increased borrowing: When a company increases its borrowing and fails to generate returns to service the debt, then the default risk faced by the creditors increases.

3. Corporate Governance Mechanisms

Corporate governance can be defined as the arrangement of checks, balances, and incentives that exists to manage conflicting interests among a company's management, board, shareholders, creditors, and other stakeholders.

A sound governance structure ensures that a corporation has mechanisms in place that not only facilitate adherence to rules and regulations imposed by external authorities, but also meet the specific needs of internal stakeholders.

Shareholder Mechanisms

Mechanisms to mitigate shareholder risks include company reporting and transparency, general meetings, investor activism, derivative lawsuits, and corporate takeovers.

Corporate Reporting and Transparency

Shareholders have access to financial and non-financial information about the company from annual reports, proxy statements, disclosures on the company's website, the investor relations department, and other channels. Such information helps shareholders to:

- reduce the extent of information asymmetry between shareholders and managers;
- assess the performance of the company and that of its directors and managers;
- make informed decisions in valuing the company and deciding to purchase, sell, or transfer shares; and
- vote on key corporate matters or changes.

Shareholder Meetings

General meetings provide an opportunity to shareholders to exercise their vote on major corporate issues. There are typically two types of general meetings:

- *Annual general meetings:* These are usually held once a year. During an AGM, a company's annual performance is presented and discussed, and shareholders' questions are answered.
- *Extraordinary general meetings:* These can be called anytime during the year, either by the company or shareholders, whenever a major resolution has to be passed such as an amendment to a company's bylaws, mergers or acquisitions, or the sale of businesses.

Number of votes required may be one of the following two types based on the type of resolution to be passed:

- For simple decisions, a simple majority of votes is sufficient.
- For material decisions, a supermajority vote is required, i.e., 75% of the votes must be in favor of a resolution to be passed.

Proxy voting allows shareholders to authorize another individual to vote on their behalf at the AGM. In *cumulative voting*, shareholders may accumulate their votes to vote for one candidate in an election that involves more than one director.

Shareholder Activism

Shareholder activism refers to strategies used by shareholders to attempt to compel a company to act in a desired manner. The primary motivation of activist shareholders is to increase shareholder value. Hedge funds are amongst the most predominant shareholder activists.

Shareholder Derivative Lawsuits

Shareholder derivative lawsuits are legal proceedings initiated by one or more shareholders against board directors, management, and/or controlling shareholders of the company. However, in many countries, the law prohibits shareholders from taking legal action through the court system.

Corporate Takeovers

Shareholders prefer corporate takeover if the management of a company underperforms. The commonly used methods for corporate takeovers are as follows:

- Proxy contest: A group attempting to take a controlling position on a company's board influences shareholders to vote for them.
- Tender offer: An offer by a group seeking to gain control to purchase a shareholder's shares.
- Hostile takeover: One company tries to acquire another company by bypassing the management and directly going to the company's shareholders.

Creditor Mechanisms

Mechanisms to mitigate creditor risks include bond indenture(s), company reporting and transparency, and committee participation.

Bond Indenture

- A bond indenture is a legal contract that describes the structure of a bond, the obligations of the company, and the rights of the bondholders.
- *Covenants* are terms in a bond indenture that specify what a bond issuer may and may not do. The goal is to reduce bondholders' exposure to risk. Affirmative covenants, also known as positive covenants, require the company to take specific actions or meet certain requirements, such as maintaining adequate levels of insurance. Restrictive covenants, also known as negative covenants, require bond issuers to not perform certain actions, such as allowing the company's liquidity level to fall below a minimum coverage ratio.

- *Collaterals* are pledged assets or financial guarantees that may be used to repay bondholders if an issuer defaults on periodic payments.

Corporate Reporting and Transparency

Creditors require the company to provide periodic financial information to monitor the risk exposure and to ensure covenants are not violated.

Creditor Committees

In some countries, official creditor committees are formed once a company files for bankruptcy. Such committees are expected to represent bondholders throughout the bankruptcy proceedings and protect bondholder interests in any restructuring or liquidation.

Board of Director and Management Mechanisms

The board of directors delegates specific functions to individual committees that, in turn, report to the board on a regular basis. The most commonly established board committees are:

Audit Committee

The key functions of the audit committee include:

- monitoring the financial reporting process;
- supervising the internal audit function, annual audit plan, and annual review; and
- appointing and interacting with an external auditor as well as implementing remediation per that auditor.

Governance Committee

The key functions of the governance committee include:

- developing and monitoring corporate governance policies and practices,
- ensuring organizational compliance and remediation with applicable laws and regulations, and
- aligning organizational structure with governance principles.

Remuneration or Compensation Committee

The key functions of the remuneration committee include:

- developing director and executive remuneration policies,
- overseeing performance policy management and evaluation, and
- setting human resources (HR) policies relating to employee compensation.

Nomination Committee

The key functions of the nomination committee include:

- overseeing the director nomination and board election process,
- identifying senior leadership candidates, and
- maintaining board composition and independence.

Risk Committee

The key functions of the risk committee include:
- determining company risk profile,
- ensuring appropriate enterprise risk management, and
- aligning corporate activities with risk appetite.

Investment Committee

The key functions of the investment committee include:
- assessing of major investment opportunities, and
- evaluating board investment recommendations.

Employee Mechanisms

Labor Laws

Standard rights of employees in any country such as hours of work, pension and retirement plans, vacation and leave, are defined in labor laws. Employees can form unions in many countries to collectively influence the management on issues they may face.

Employment Contracts

Individual employee contracts define an employee's rights and responsibilities, remunerations, and other benefits such as ESOPs.

Customer and Supplier Mechanisms

Companies enter into contracts with both customers and suppliers that define the products, services, any guarantee, after-sales support, payment terms, etc. It also defines the course of action in case one party violates the contract.

Government Mechanisms

Regulations

Governments and regulatory agencies pass laws to protect the interests of consumers or specific stakeholders. Sensitive industries such as banks, health care, and food manufacturing companies have to comply with a rigorous regulatory framework.

Corporate Governance Codes

Many regulatory authorities have also adopted corporate governance codes, which are guiding principles for publicly traded companies. These codes require companies to disclose whether they have implemented recommended corporate governance practices or explain why they have not done so.

4. Corporate Governance Risks and Benefits

Risks of Poor Governance and Stakeholder Management

In this section, we analyze the risks a company faces due to a poor governance structure.

Weak Control Systems

Weak control systems and poor monitoring can affect a company's performance and value. One example is that of Enron where auditors failed to uncover fraudulent reporting that ultimately affected many stakeholders.

Ineffective Decision-Making

Poor decisions include managers avoiding good investment opportunities to maintain a low-risk profile or taking on excessive risk without properly evaluating potential investments. Both decisions are not in the interests of shareholders. Such decisions may result from information asymmetry, when managers have access to more information than the board/shareholders.

Legal, Regulatory, and Reputational Risks

Improper implementation and monitoring of corporate governance procedures may result in the following risks:

- Legal: Stakeholders such as shareholders, creditors and employees may file lawsuits against the company if their rights are violated.
- Regulatory: Government/regulator may choose to take action if the applicable laws are violated.
- Reputational: A company may be subjected to negative publicity by investors/analysts if there is an improperly managed conflict of interest.

Default and Bankruptcy Risks

Poor corporate governance may affect the company's performance, which in turn may affect the company's ability to service its debt. If creditors' rights are violated and they choose to take legal action on defaulting debt, the company may be forced to file for bankruptcy.

Benefits of Effective Governance and Stakeholder Management

The benefits of good corporate governance and stakeholder management are as follows:

- Operational efficiency: A good governance structure ensures that all employees of a company have a clear understanding of their responsibilities and reporting structures. The operational efficiency of a company is improved when good governance structure is combined with strong internal control mechanisms.

- Improved control: Good governance implies improved control at all levels to help a company manage its risk efficiently. Control can be improved with a good audit committee, complying with laws and regulations, and introducing procedures to handle related-party transactions.
- Better operating and financial performance: A company's operating and financial performance can be improved with good governance practices. Proper remuneration for management, mitigation of lawsuits against the company, and improving the decision-making of its managers to make the right investments are ways that will help in improving the performance of a company.
- Lower default risk and cost of debt: A company's cost of debt and default risk can be reduced by protecting creditors' rights, ensuring proper audits are conducted, and that there is no information gap between the company and its creditors.

Summary

LO: Describe the principal-agent relationship and conflicts that may arise between stakeholder groups.

A principal-agent relationship arises when a principal hires an agent to carry out a task or a service. An agent is obliged to act in the best interests of the principal and should not have a conflict of interest in performing a task. However, such relationships often lead to conflicts among stakeholders in a corporate structure. Examples of relationships that lead to such conflicts include:

- shareholder and manager/director.
- controlling and minority shareholder.
- shareholder and creditor.

LO: Describe corporate governance and mechanisms to manage stakeholder relationships and mitigate associated risks.

Corporate governance can be defined as the arrangement of checks, balances, and incentives that exists to manage conflicting interests among a company's management, board, shareholders, creditors, and other stakeholders.

Mechanisms to mitigate shareholder risks include company reporting and transparency, general meetings, investor activism, derivative lawsuits, and corporate takeovers.

Mechanisms to mitigate creditor risks include bond indenture(s), company reporting and transparency, and committee participation.

Mechanisms to mitigate board risks include board/management meetings and board committees.

Mechanisms to mitigate risks for other stakeholder (employees, customers, suppliers, and regulators) include policies, laws, regulations, and codes.

LO: Describe potential risks of poor corporate governance and stakeholder management and benefits of effective corporate governance and stakeholder management.

The risks of poor corporate governance include weak control systems; ineffective decision-making; legal, regulatory, and reputational risks; and default and bankruptcy risks. Benefits include operational efficiency, improved control, better operating and financial performance, and lower default risk and cost of debt.

LM04 Working Capital and Liquidity

Ver 1.0

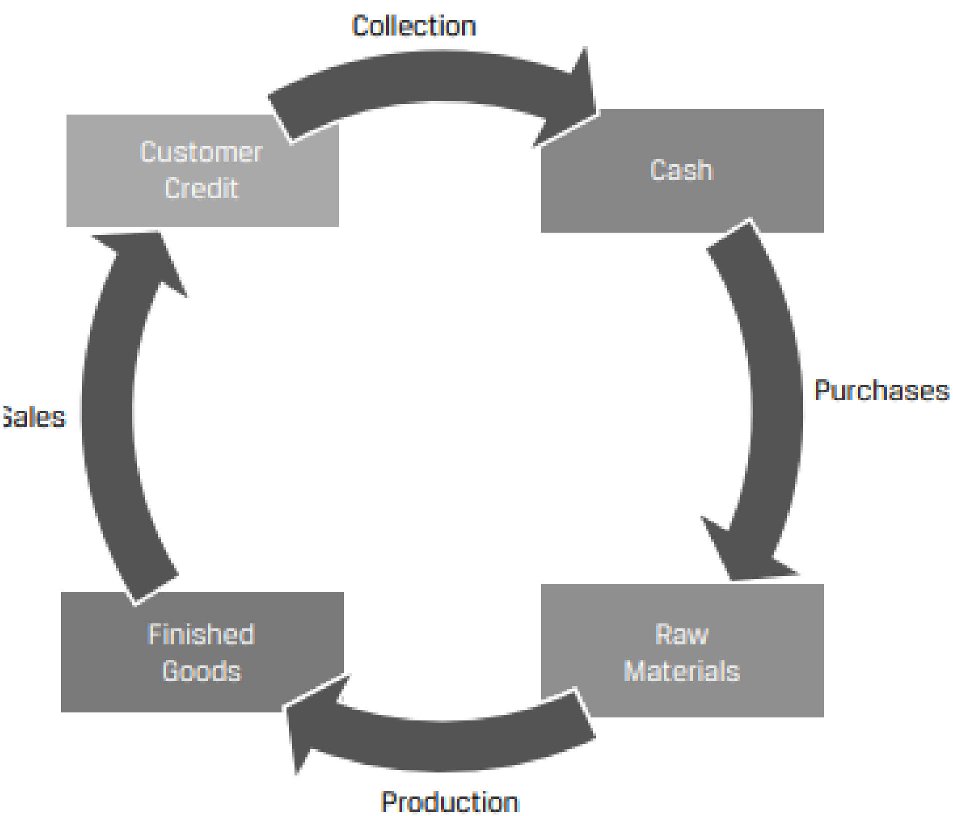
1. Introduction

This learning module covers:

- Cash conversion cycle
- Primary and secondary sources of liquidity; evaluating a company's liquidity position
- Working capital approaches and their impact on the funding needs of a company

2. Cash Conversion Cycle

The business operations of a company are typically made up of several sequential steps such as: acquiring raw materials, producing inventory, selling products to customers, and collecting cash. These activities are known as the issuer's operating cycle, and they can occur once or several times per year. Exhibit 1 from the curriculum illustrates an operating cycle.



These activities generate cash outflows and inflows that do not always occur at the same time as the activity. For example, raw materials may be purchased on credit and a payment made to the vendor later. Finished goods can be sold and delivered to customers but payment collected later. Inventory may take time to produce and be ready for sale.

Future cash inflows from operating activities are recorded as short-term assets, while future cash outflows are recorded as short-term liabilities.

Exhibit 2 from the curriculum defines a few short-term assets and liabilities that are of interest to analysts.

Short-Term Asset	**Meaning**
Accounts receivable	Amounts to be collected from customers for products or services sold
Inventory	Cost of products produced or purchased for sale
Short-Term Liability	**Meaning**
Accounts payable	Amounts owed to suppliers for products or services received

Exhibit 3 from the curriculum demonstrates when these short-term assets and liabilities are recognized and de-recognized on the balance sheet.

Short-Term Asset	**Recognized When . . .**	**Derecognized When . . .**
Accounts receivable	Product or service is sold to customer on credit	Cash is received from customer
Inventory	Issuer takes ownership of materials, goods, supplies, etc.	Product is sold to customer
Short-Term Liability	**Recognized When . . .**	**Derecognized When . . .**
Accounts payable	Product or service is received, and issuer defers payment to supplier	Cash is paid to supplier

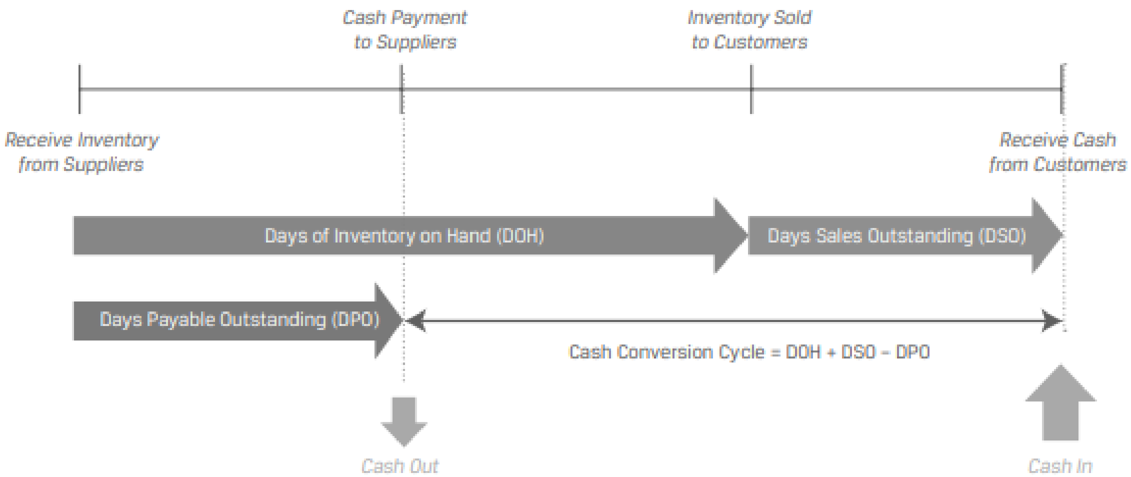
The amounts of time that accounts payable, inventory, and accounts receivable are outstanding on the balance sheet are known, respectively, as days payable outstanding (DPO), days of inventory on hand (DOH), and days sales outstanding (DSO).

Instructor's Note: The calculation of these amounts will be covered in a later learning module.

Cash Conversion Cycle: A company's cash conversion cycle is the amount of time between an issuer paying its suppliers and receiving cash from customers. (i.e., the time between derecognition of accounts payable and derecognition of accounts receivable)

Cash conversion cycle = Days of inventory on hand + Days sales outstanding – Days payable outstanding.

Exhibit 4 form the curriculum illustrates a cash conversion cycle graphically.



Generally, the shorter the cycle the better. A short cash conversion cycle means that a company converts its inventory investment into cash quickly, whereas a long cash conversion cycle means that a company converts its inventory investment into cash slowly. The longer the cycle the higher the amount of working capital a company will need.

Some companies can also have a negative cash conversion cycle – they receive cash from customers well before suppliers are paid.

Example: Cash Conversion for US Retailers
(*This is based on Example 2 from the curriculum.*)

The following information is provided for five large US discount retailers Walmart Inc., Target Corporation, Costco Wholesale Corporation, The TJX Companies, and Ross Stores for the 2021 calendar year.

	Walmart	**Target**	**Costco**	**TJX**	**Ross**
Days sales outstanding	5	2	3	7	2
Days of inventory on hand	48	68	29	63	61
Days payable outstanding	47	75	34	47	63

Which company has the shortest cash conversion cycle?

Solution:

Cash conversion cycle = DSO + DOH − DPO

Cash conversion cycle for Walmart = 5 days + 48 days − 47 days = 6 days

Cash conversion cycle for Target = 2 days + 68 days −75 days = −5 days

Cash conversion cycle for Costco = 3 days + 29 days − 34 days = −2 days

Cash conversion cycle for TJX = 7 days + 63 days − 47 days = 23 days

Cash conversion cycle for Ross = 2 days + 61 days − 63 days = 0 days

Target has the shortest cash conversion cycle of the five companies.

Companies can shorten their cash conversion cycle in several ways:

- Reduce days on inventory on hand by discounting products with low demand, using a 'just in time' inventory system, using data analytics to improve demand forecasts.
- Reduce days sales outstanding by offering prompt-payment discounts to customers, imposing late fees, working with third-party collection agencies.
- Increase days payable outstanding by negotiating longer payment terms with suppliers. However, the ability to negotiate better terms is highly dependent on the power dynamics between the company and its supplier. A small company purchasing a specialized critical component from a sole supplier, may not have the ability to negotiate better terms.

Suppliers usually offer discounts for prompt payments. For example, the terms may be: 2/10 net 30, which means if the payment is made within 10 days, the company will get a 2 percent discount else the entire payment must be made within 30 days. In such cases the company should compare the cost of trade credit with its other short-term borrowing costs to make a decision on whether to take the discount.

Cost of trade credit = $\left(1 + \frac{\text{discount}}{1-\text{discount}}\right)^{n} - 1$
where: n = 365 / number of days beyond discount period

This is illustrated in the example below:

Example: Internal versus External Financing Decision
(*This is based on Example 1 from the curriculum.*)

A company is evaluating two options to fund its working capital needs:

- Option1: Forgo the 2% discount offered by its supplier. The standard trade terms are 2/10 net 30.
- Option 2: Borrow through its external line of credit. The effective annual rate for the line of credit is 7.7%.

Which option should the company prefer?

Solution:

The effective annual rate on the forgone trade credit can be calculated as:

[(1+(2/98))^(365/20)] – 1 = 44.6%

This is significantly higher than the 7.7% rate on the external credit line. Therefore, the company should prefer the credit line. It should borrow from the credit line and pay the supplier early to avail the discount.

Apart from the cash conversion cycle, analysts also assess the amount of working capital used by a company.

Total working capital = Current assets – Current Liabilities

Net working capital = Current assets, excluding cash and marketable securities – Current liabilities, excluding short-term and current debt

To compare across firms, total or net working capital is often expressed as a percentage of sales. The lower this ratio, the better.

Example: Net Working Capital
(*This is based on Question 6 from the curriculum.*)

Consider the following balance sheet for an issuer.

Cash	100
Marketable securities	20
Accounts receivable	600
Inventory	800
Prepaid expenses	30
Property, plant, and equipment	10,000
Intangibles	500
Total assets	12,050
Accounts payable	980
Accrued expenses	70
Short-term debt	1,000
Long-term debt	2,000
Shareholders' equity	8,000
Total liabilities and equity	12,050

Calculate the issuer's net working capital.

Solution:

Net working capital = Current assets, excluding cash and marketable securities – Current liabilities, excluding short-term and current debt

Current assets, excluding cash and marketable securities:

Accounts receivable	600
Inventory	800
Prepaid expenses	30
Sum	1,430

Current liabilities, excluding short-term and current debt:

Accounts payable	980

Accrued expenses	70
Sum	1,050

Net working capital = 1,430 – 1,050 = 380

3. Liquidity

'Liquidity' is the extent to which a company is able to meet its short-term obligations using cash flows and those assets that can be readily transformed into cash.

The liquidity of an asset can be evaluated along two dimensions:

- The type of the asset
- The speed at which the asset can be converted into cash (by sale or financing)

'Liquidity management' refers to the company's ability to generate cash when needed, at the lowest possible cost.

Two sources of liquidity for a company are:

- primary sources of liquidity, such as cash balances
- secondary sources of liquidity, such as selling assets

The main difference between the two is that using primary sources has no effect on a company's ongoing operations, whereas using secondary sources may have a negative impact on a company's ongoing operations.

Primary Liquidity Sources

They represent the most readily accessible resources available to the company. Primary sources include:

- <u>Free cash flow</u>: The firm's after-tax operating cash flow less planned short- and long-term investments.
- <u>Cash and marketable securities on hand</u>: Cash available in bank accounts or held as marketable securities that can be sold quickly without loss of value.
- <u>Borrowings</u> from banks, bondholders, or supplier's trade credit. This option settles the current obligation but creates a new obligation that has to be repaid in future.

Secondary Liquidity Sources

Secondary sources include:

- Suspending or reducing dividend payments.
- Delaying or reducing capital expenditures
- Issuing additional equity
- Renegotiating debt contracts
- Selling assets

- Filing for bankruptcy protection and reorganization

Example: Estimating Costs of Liquidity
(*This is based on Example 3 from the curriculum.*)

A company facing a liquidity crisis has the following options to raise funds. The estimated fair value and liquidation costs for each source of funds is listed below:

Source of Funds	Fair Value (C$, millions)	Liquidation Costs (%)
Sell short-term marketable securities	10	0
Sell select inventories and receivables	20	10
Sell excess real estate property	50	15
Sell a subsidiary of the firm	30	20

Net of liquidation costs, how much liquidity can the company raise if all four sources of funds are used?

Solution:

The costs and net funds raised are summarized in this table:

Source of Funds	Fair Value (C$, millions)	Liquidation Costs (%)	Liquidation Costs (C$, millions)	Net Proceeds (C$, millions)
Marketable securities	10	0	0	10
Inventories and receivables	20	10	2	18
Real estate property	50	15	7.5	42.5
Subsidiary of the firm	30	20	6	24
Total			15.5	94.5

Factors Affecting Liquidity: Drags and Pulls

A company's liquidity position is affected by cash receipts and the amount of cash it has to pay.

'Drags on liquidity' reduce cash inflows. For example, uncollected receivables, obsolete inventory, tight credit etc.

'Pulls on liquidity' accelerate cash outflows. For example, earlier payment of vendor dues, reduced credit limits (by suppliers), limits on short-term lines of credit (by banks) etc.

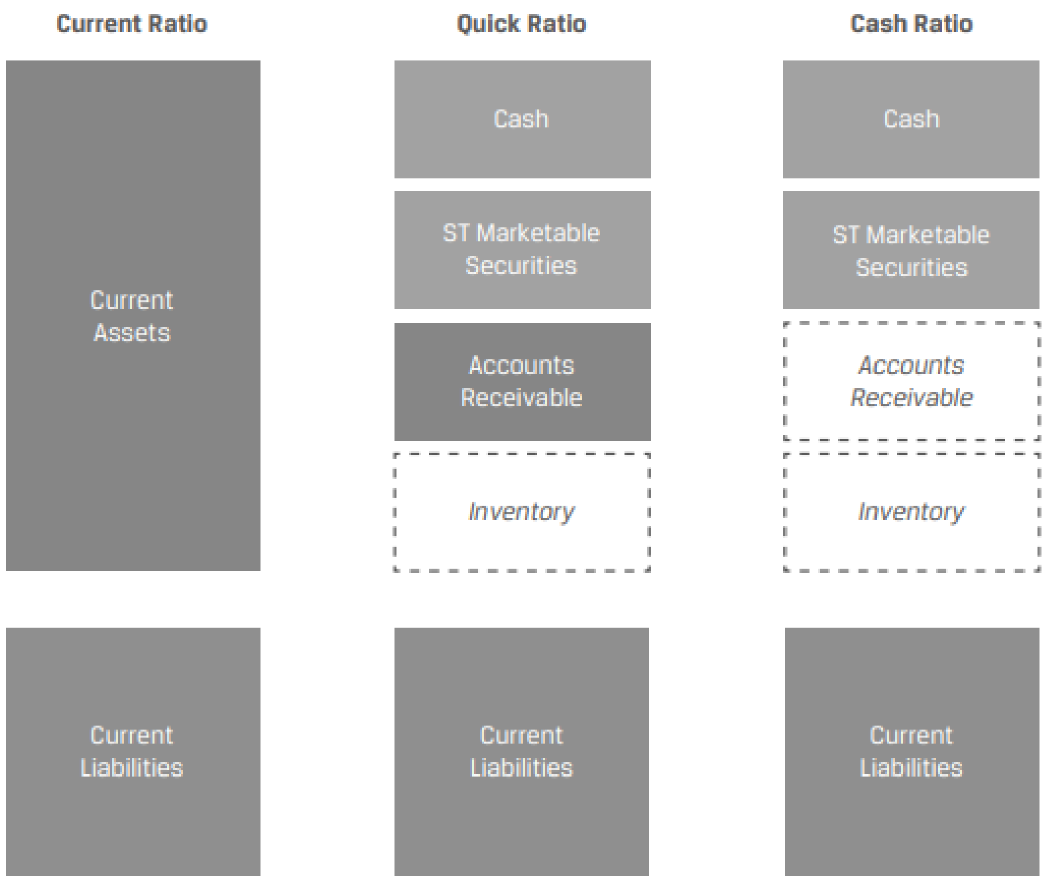
Measuring and Evaluating Liquidity

Liquidity ratios are used to measure a company's ability to meet its short-term obligations.

The three commonly used liquidity ratios are:

Ratio	Formula	Interpretation
Current ratio	Current assets ÷ Current liabilities	A higher number implies greater liquidity.
Quick ratio	(Cash + Marketable securities + Receivables) ÷ Current liabilities	A higher number implies greater liquidity. More conservative than current ratio as inventory is excluded.
Cash ratio	(Cash + Marketable securities) ÷ Current liabilities	This is the most conservative liquidity ratio and a good measure of a company's ability to handle a crisis situation.

The denominator is the same for all ratios. Exhibit 9 from the curriculum illustrates what is included in the numerator for each ratio.



The levels of these ratios, and their trends, or changes over time, in addition to comparisons with competitors or the industry, are used to judge a firm's liquidity position.

4. Managing Working Capital and Liquidity

Working Capital Management

To determine their required working capital investment, companies first identify their optimal levels of inventory, receivables and payables as a function of sales. They then project these assumptions forward into the future.

After determining its working capital requirements, a firm then identifies the optimal mix of short-and long-term financing necessary to fund these requirements. Companies may take different approaches to working capital management:

- Conservative approach: In a conservative approach, the firm holds a larger position in current assets (cash, receivables, and inventories). This provides financial flexibility to respond to unforeseen events, but it results in a lower return on equity.

  Instructor's Note: Current assets can be divided into 'permanent current assets' and 'variable current assets.' Permanent current assets remain relatively constant throughout the year. Variable current assets vary based on the seasonality of the business, increasing during peak production periods.

  In a conservative approach, the firm finances a majority of its current assets (both permanent and variable) with long-term debt or equity financing.

- Aggressive approach: In an aggressive approach, the firm holds a substantially smaller position in current assets. This reduces financial flexibility, but it results in a higher return on equity.

  Also, the firm finances a majority of its current assets (both permanent and variable) with short-term debt or payables.

- Moderate approach: In a moderate approach, the firm holds a position in current assets that is somewhere between the two approaches.

  Also, the firm attempts to match the duration of the current assets with the liabilities. It would finance its permanent current assets with long-term debt and equity; and finance its variable current assets with short-term debt and payables.

Exhibits 11, 12, and 13 from the curriculum presents the pros and cons of each approach.

Conservative approach

Pros	*Cons*
Stable, permanent financing avoids rollover risk associated with short-term debt	Long-term debt typically involves a higher interest rate
Financing costs are known upfront	High cost of equity

Certainty of working capital needed to purchase the necessary inventory	Permanent financing eliminates the opportunity to borrow only as needed (increasing ongoing financing costs)
Extended payment term reduces short-term cash needs for debt service	A longer lead time is required to establish the financing position
Higher flexibility during market disruptions that can be covered by larger cash or marketable securities positions	Long-term debt may require more covenants that restrict business operations

<u>Aggressive approach</u>

Pros	*Cons*
Short-term lines of credit provide the flexibility to access financing only when needed—particularly appropriate for seasonality—reducing overall interest expense	Interest expense may fluctuate as rates on short-term financing change
Short-term loans involve less rigorous credit analysis, as the lender has greater clarity as to the short-term operations of the firm	May result in higher short-term cash needs to satisfy debt maturities
Flexibility to refinance if rates decline	Rollover risk of short-term debt increases bankruptcy risk, particularly during market disruptions
	May have to rely on more costly trade credit, tighten customer credit, or sell receivables if unable to refinance at favorable terms

<u>Moderate approach</u>

Pros	*Cons*
Lower financing cost versus conservative approach; lower risk than aggressive approach	Access to short-term capital may be limited for seasonal or growth needs
Flexibility to increase financing for seasonal spikes while maintaining a base level for ongoing needs	Uncertain cost of short-term debt for variable needs during market disruptions

Diversifying sources of funding with a more disciplined approach to balance sheet management	May have to rely on more costly trade credit to meet seasonal or growth needs if unable to refinance at favorable terms

Liquidity and Short-Term Funding

Companies seek to implement a short-term financing strategy that will help achieve the following objectives:

- Ensure sufficient capacity to handle peak cash needs.
- Maintain sufficient and diversified sources of credit.
- Ensure rates are cost-effective.
- Ensure both implicit and explicit funding costs are considered.

Factors that will influence a company's short-term borrowing strategies are:

- Size: Larger companies have additional and cheaper options as compared to smaller companies.
- Creditworthiness: The borrower's creditworthiness determines the access to and the cost of borrowing.
- Legal and regulatory considerations: Legal and regulatory constraints on specific industries can restrict how much a company can borrow and under what terms.
- Underlying assets: Depending on their business model, some companies may have assets that are considered attractive as collateral for secured loans.

Summary

LO: Explain the cash conversion cycle and compare issuers' cash conversion cycles.

A company's cash conversion cycle is the amount of time between an issuer paying its suppliers and receiving cash from customers.

Cash conversion cycle = Days of inventory on hand + Days sales outstanding – Days payable outstanding.

Companies can shorten their cash conversion cycle in several ways:

- Reduce days on inventory on hand by discounting products with low demand, using a 'just in time' inventory system, using data analytics to improve demand forecasts.
- Reduce days sales outstanding by offering prompt-payment discounts to customers, imposing late fees, working with third-party collection agencies.
- Increase days payable outstanding by negotiating longer payment terms with suppliers

LO: Explain liquidity and compare issuers' liquidity levels.

'Liquidity' is the extent to which a company is able to meet its short-term obligations using cash flows and those assets that can be readily transformed into cash.

The three commonly used liquidity ratios are:

Ratio	*Formula*	*Interpretation*
Current ratio	Current assets ÷ Current liabilities	A higher number implies greater liquidity.
Quick ratio	(Cash + Marketable securities + Receivables) ÷ Current liabilities	A higher number implies greater liquidity. More conservative than current ratio as inventory is excluded.
Cash ratio	(Cash + Marketable securities) ÷ Current liabilities	This is the most conservative liquidity ratio and a good measure of a company's ability to handle a crisis situation.

LO: Describe issuers' objectives and compare methods for managing working capital and liquidity.

Issuers can take a conservative, moderate, or aggressive approach to working capital management. These approaches differ in the amount of working capital held on the balance sheet, their reliance on external financing, and the composition of short- and long-term financing.

Companies seek to implement a short-term financing strategy that will help achieve the following objectives:

- Ensure sufficient capacity to handle peak cash needs.
- Maintain sufficient and diversified sources of credit.
- Ensure rates are cost-effective.
- Ensure both implicit and explicit funding costs are considered.

LM05 Capital Investments and Capital Allocation

Ver 1.0

1. Introduction

Capital investments are investments with a life of one year or more. Companies make capital investments to generate value for their shareholders.

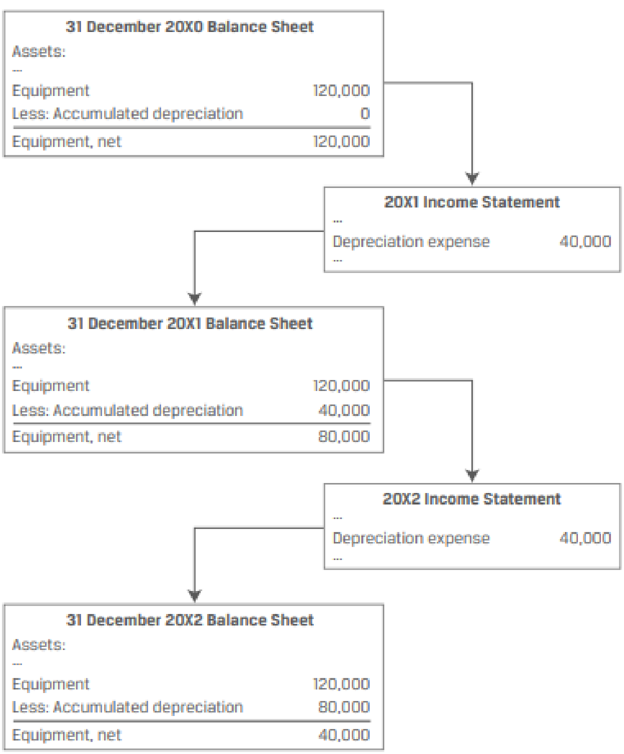
Capital allocation is the process by which companies make capital investment decisions. Capital allocation is important because it impacts a company's future.

This learning module covers:

- Types of capital investments – going concern, regulatory/compliance, expansion, and other projects
- Steps in a typical capital allocation process
- Basic investment decision criteria – NPV, IRR, and ROIC
- Principles of capital allocation and common capital allocation pitfalls
- Types of real options – timing, sizing, flexibility, and fundamental options

2. Capital Investments

Capital investments are shown on the balance sheet as long-term assets. A portion of the cost is recorded on the income statement periodically as a non-cash depreciation or amortization expense over the asset's useful life. In subsequent periods, the amount on the balance sheet is shown on a net basis, i.e., initial cost – accumulated depreciation. The net value declines to zero or a salvage value at the end of the asset's useful life. Exhibit 1 from the curriculum illustrates this process.



Types of Capital Projects

Companies invest for two primary reasons – to maintain their existing business and to grow it.

Projects undertaken by companies to maintain the business include:

- Going concern projects: Projects necessary to continue current operations and maintain existing size of the business or to improve business efficiencies.
  Example: machine replacement, infrastructure improvement
- Regulatory/Compliance projects: Projects typically required by a third party, such as the government regulatory body, to meet specified safety and compliance standards.
  Example: factory pollution control installation, performance bond posting to guarantee satisfactory project completion

Projects undertaken by companies to expand the business include:

- Expansion projects: Projects that expand business size and typically involve greater degrees of risk and uncertainty than going concern projects.
  Example: new product or service development, merger, acquisition
- Other projects: Projects, which include high-risk investments and new growth initiatives, that are outside the company's conventional business lines.
  Example: exploration investment into a new innovation, business model, or idea

Going Concern Projects

- Projects necessary to continue current operations and maintain existing size of the business or to improve business efficiencies.
- Most common going concern projects are replacing assets that have reached the end of their useful life, and maintaining IT hardware and software.
- Typically, these projects do not increase revenues but they could help a company save costs.
- They are fairly easy to evaluate as compared to the other projects.
- To fund these projects, managers often try to match the financing with the life-span of the asset. For example, issuing a 20-year bond to fund an asset with an expected life of 20-years.
  - If a company finances long-term assets with short-term financing, it faces rollover risk – the short-term financing costs could go up.
  - If a company finances short-term assets with long-term financing, it faces the risk of overpaying in financing costs.
- To estimate the amount of capital spending on going concern projects, analysts often look at the depreciation and amortization expense reported on the income statement.

Regulatory Compliance Projects

- Projects typically required by a third party, such as the government regulatory body, to meet specified safety and compliance standards.
- These projects are unlikely to increase revenue; in fact, they may increase compliance costs. However, regulatory/compliance costs can act as barriers to entry and protect the industry's profitability.
- While evaluating these projects, management must determine whether the business will still be profitable after considering the additional compliance costs.
  - In most cases, companies will accept these projects and pass on the additional cost to customers.
  - However, sometimes the cost may be too high and the company may decide that it is better off ceasing operations or shutting down the part of the business that is subject to the new regulation.

Expansion of Existing Business

- Projects that expand business size and typically involve greater degrees of risk and uncertainty than going concern projects.
- Some industries such as pharmaceuticals and oil and gas spend heavily (over 10% of revenues) on expansion projects.
- If internal opportunities for expansion are limited, managers may pursue acquisitions. However, the two main risks in acquisitions are the risk of overpaying and the difficulty in integrating the acquirer's business operations.

New Lines of Business and Other Projects

- Projects, which include high-risk investments and new growth initiatives, that are outside the company's conventional business lines.
- These projects are often observed in privately held companies or public companies under the control of a founding owner/significant shareholder.
- These projects tend to have a venture capital element to them. There will probably be a complete loss of investment, but if successful the project could be highly profitable.

3. Capital Allocation

Capital allocation is the process used by an issuer's management to make capital investment decisions.

Steps in Capital Allocation Process

The steps in the capital allocation process are as follows:

Step 1 – Idea generation: Most important step in the process. Investment ideas can come from anywhere within the organization, or outside (customers, vendors, etc.). What projects can add value to the company in the long term?

Step 2 – Investment analysis: Gathering information to forecast cash flows for each project and then computing the project's profitability. Output of this step: A list of profitable projects.

Step 3 – Planning and prioritization: Do the profitable projects fit in with the company's long-term strategy? Is the timing appropriate? Some projects may be profitable in isolation but not so much when considered along with the other projects. Scheduling and prioritizing of projects are important.

Step 4 - Monitoring and post-investment review: Post-investment review helps in assessing how effective the capital budgeting process was. How do the actual revenues, expenses, and cash flows compare against the predictions? Post-investment review is useful in three ways:

- If the predictions were optimistic or too conservative, then it becomes evident here.
- Helps improve business operations. Puts the focus on out-of-line sales and costs.
- Helps in identifying profitable areas for fresh investments in the future, or scale down in non-profitable ones.

Two widely used analytical tools in the 'investment analysis' step are NPV and IRR.

Net Present Value (NPV)

Net present value is the present value of the future after tax cash flows minus the investment outlay.

$$\mathrm{NPV} = \mathrm{CF0} + \left[\frac{\mathrm{CF1}}{(1+\mathrm{r})^1}\right] + \left[\frac{\mathrm{CF2}}{(1+\mathrm{r})^2}\right] + \left[\frac{\mathrm{CF3}}{(1+\mathrm{r})^3}\right]$$

Decision rule:

For independent projects:

If NPV > 0, accept.

If NPV < 0, reject.

For mutually exclusive projects:

Accept the project with higher and positive NPV.

Example

Compute NPV for projects A and B given the following data:

Cost of capital = 10%		
Expected Net after Tax cash flows		
Year	**Project A (in $)**	**Project B (in $)**

0	-1,000	-1,000
1	500	100
2	400	300
3	300	400
4	100	600

Solution:

Project A:

NPV = -1000 + $\frac{500}{1.1}$ + $\frac{400}{1.1^2}$ + $\frac{300}{1.1^3}$ + $\frac{100}{1.1^4}$ = 78.82

On the exam, you can save time by using the calculator to solve for NPV instead of using the above formula. The key strokes are given below:

Key strokes	Display
[CF][2nd][CLR WORK]	CF0 = 0
1000 [+|-] [ENTER]	CF0 = -1000
[↓] 500 [ENTER]	C01 = 500
[↓]	F01 = 1
[↓] 400 [ENTER]	C02 = 400
[↓]	F02 = 1
[↓] 300 [ENTER]	C03 = 300
[↓]	F03 = 1
[↓] 100 [ENTER]	C04 = 100
[↓]	F04 = 1
[NPV] 10 [ENTER]	I = 10
[↓] CPT	NPV = 78.82

Project B:

NPV = -1000 + $\frac{100}{1.1}$ + $\frac{300}{1.1^2}$ + $\frac{400}{1.1^3}$ + $\frac{600}{1.1^4}$ = 49.18

Microsoft Excel functions can also be used to solve for the NPV. The two functions available are:

- NPV or =NPV(rate, values) and
- XNPV or =XNPV(rate, values, dates),

where "rate" is the discount rate, "values" are the cash flows, and "dates" are the dates of each of the cash flows.

Internal Rate of Return (IRR)

IRR is the discount rate that makes the present value of future cash flows equal to the investment outlay. We can also say that IRR is the discount rate which makes NPV equal to 0.

<u>Decision rule</u>:

For independent projects:

If IRR > required rate of return (usually firms cost of capital adjusted for projects riskiness), accept the project.

If IRR < required rate of return, reject the project.

The required rate of return is also called 'hurdle rate'**.**

For mutually exclusive projects:

Accept the project with higher IRR (as long as IRR > cost of capital).

Example

Compute IRR for projects A and B given the following data.

Cost of Capital = 10%; Expected Net After Tax Cash Flows		
Year	**Project A (in $)**	**Project B (in $)**
0	-1,000	-1,000
1	500	100
2	400	300
3	300	400
4	100	600

Solution:

Project A:

A very tedious method is to set up the equation below and solve for r using trial and error.

$$1000 = \frac{500}{1+r} + \frac{400}{(1+r)^2} + \frac{300}{(1+r)^3} + \frac{100}{(1+r)^4} \qquad r = 14.49\%$$

A much faster method is to use the calculator:

Key strokes	**Display**
[CF][2^{nd}][CLR WORK]	CF0 = 0
1000[+|-] [ENTER]	CF0 = -1000
[↓] 500 [ENTER]	C01 = 500
[↓]	F01 = 1
[↓] 400 [ENTER]	C02 = 400
[↓]	F02 = 1
[↓] 300 [ENTER]	C03 = 300
[↓]	F03 = 1
[↓] 100 [ENTER]	C04 = 100
[↓]	F04 = 1
[IRR][CPT]	14.49

Project B:

$$1000 = \frac{100}{1+r} + \frac{300}{(1+r)^2} + \frac{400}{(1+r)^3} + \frac{600}{(1+r)^4} \qquad r = 11.79\%$$

Microsoft Excel functions can also be used to solve for the IRR. The two functions available are:

- IRR or =IRR(values, guess)
- XIRR or =XIRR(values, dates, guess)

where "values" are the cash flows, "guess" is an optional user-specified guess that defaults to 10%, and "dates" are the dates of each cash flow.

Ranking conflicts between NPV and IRR

For single and independent projects with conventional cash flows, there is no conflict between NPV and IRR decision rules. However, for mutually exclusive projects the two criteria may give conflicting results. The reason for conflict is due to differences in cash flow patterns and differences in project scale.

For example, consider two projects one with an initial outlay of $1 million and another project with an initial outlay of $1 billion. It is possible that the smaller project has a higher IRR, but the increase in firm value (NPV) is small as compared to the increase in firm value (NPV) of the larger project.

In case of a conflict, we should always go with the NPV criterion because:

- The NPV is a direct measure of expected increase in value of the firm.
- The NPV assumes reinvestment of cash flows at the required rate of return (more realistic), whereas the IRR assumes reinvestment of cash flows at the IRR rate (less realistic).
- IRR is not useful for projects with non-conventional cash flows as such projects can have multiple IRRs , i.e., there are more than one discount rates that will produce an NPV equal to zero.

Comparison between NPV and IRR

NPV	*IRR*
Advantages	Advantages
Direct measure of expected increase in value of the firm.	Shows the return on each dollar invested.
Theoretically the best method.	Allows us to compare return with the required rate.
Disadvantages	Disadvantages
Does not consider project size.	Incorrectly assumes that cash flows are reinvested at IRR rate. The correct assumption is that intermediate cash flows are reinvested at the required rate.
	Might conflict with NPV analysis.

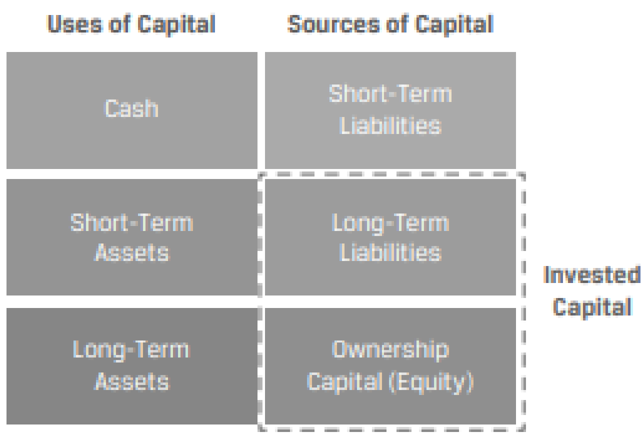
	Possibility of multiple IRRs.

Return on Invested Capital

Outside investors and analysts often do not have the information necessary to calculate a project's NPV and IRR. They get consolidated financial statements which includes the cash flows associated with many projects. Therefore, they use a profitability measure - return on invested capital to evaluate the capital allocation decisions of a company.

$$\text{ROIC} = \frac{\text{After tax operating profit}}{\text{Average invested capital}}$$

Invested capital includes long-term liabilities and equity, short-term liabilities are excluded.



ROIC reflects how effectively a company's management is able to convert capital into after-tax operating profits. Note that the numerator excludes interest expense because it represents a source of return to providers of debt capital, and the denominator includes sources of capital from all providers.

If the ROIC measure is higher than the cost of capital (COC), the company is generating a higher return for investors compared with the required return, thereby increasing the firm's value. We can also say that projects with positive NPV will have a ROIC that is greater than the COC.

Example: Return on Invested Capital
(*This is based on Example 8 from the curriculum.*)

Assume that a company reported 24,395 in Year 2 after-tax operating profits and the following balance sheet information.

Assets	End of Year 1	End of Year 2
Cash	4,364	6,802
Short term assets	40,529	52,352
Long term assets	287,857	279,769

Total assets	332,750	338,923

Liabilities and Equity:	End of Year 1	End of Year 2
Accounts payable	35,221	50,766
Short-term debt	21,142	5,877
Long-term debt	112,257	106,597
Share capital	15,688	15,688
Retained earning	148,442	159,995
Total liabilities and equity	332,750	338,923

Calculate the ROIC for Year 2.

Solution:

$$\text{ROIC} = \frac{\text{After tax operating profit}}{\text{Average invested capital}} = \frac{\text{After tax operating profit}}{\text{Average LT liabilities and equity}}$$

$$\text{ROIC} = (24{,}395) / \frac{(112{,}257 + 15{,}688 + 148{,}442) + (106{,}597 + 15{,}688 + 159{,}995)}{2} = 8.73\%$$

4. Capital Allocation Principles and Pitfalls

Capital Allocation Principles

The key principles of capital allocation are:

- Decisions are based on after-tax cash flows: Decisions are not based on an accounting/accrual basis (net income or operating income) which subtract non-cash charges such as depreciation. Shareholder value increases only on the cash that they have earned. Hence, any tax expenses must be deducted from the cash flows.
- Measure incremental cash flows: What incremental cash flows occur with the investment as compared to without it? Also sunk costs are excluded, while externalities are included.
  - Exclude sunk costs: A sunk cost is one that has already been incurred and cannot be changed. For example, already incurred costs like preliminary consulting fees should not be included in the analysis
  - Include externalities: Both positive/negative externalities should be considered in the analysis. Cannibalization is an example of a negative externality. For example, negative impact of a new diet soda product launch on the sales of existing soda products.
- Timing of cash flows is crucial: Due to the time value of money, cash flows received earlier are more valuable than cash flows received later.

Capital Allocation Pitfalls

Common capital allocation pitfalls can be divided into cognitive errors and behavioral biases. Cognitive errors include calculation and other mistakes, while behavioral biases include errors in judgment and blind spots.

Cognitive Errors in Capital Allocation

- *Internal forecasting errors*: Companies may make errors in their internal forecasts. The errors could be related to incorrect costs or discount rate inputs. For example, overhead costs such as management time, IT support, and financial systems can be difficult to estimate. Also, companies may incorrectly treat sunk costs and missed opportunity costs; and incorrectly use the company's overall cost of capital, cost of debt, or cost of equity rather than the investment's required rate of return in their analysis.
- *Ignoring costs of internal financing*: Many managers consider internally generated capital to be "free" and allocate it according to a budget that is heavily correlated with prior period amounts. Externally raised capital, on the other hand, is regarded as "expensive" and is used sparingly, typically only for large investments. Ideally, managers should view all capital as having an opportunity cost, regardless of source.
- *Inconsistent treatment of, or ignoring inflation*: Capital allocation analysis can be done either in 'nominal' terms or 'real' terms. Nominal cash flows include the effects of inflation. Whereas, real cash flows are adjusted downward to remove the effect of inflation. Nominal cash flows should be discounted at a nominal discount rate, and real cash flows should be discounted at a real rate.

Behavioral Biases in Capital Allocation

- *Inertia:* The amount of capital investment in a business segment/unit for a year is highly correlated to the amount spent in the previous year. Ideally, the amount should vary based on the number and scale of opportunities available each year.
- *Basing investment decisions on EPS, net income, or return on equity*: The compensation of managers is sometimes tied to EPS, net income, or ROE. They may therefore reject even strong positive NPV projects that reduce these accounting numbers in the short run.
- *Pushing "pet" projects*: Pet projects are projects backed by senior management. They may contain overly optimistic projections that overstate the project's profitability.
- *Failing to consider investment alternatives or alternative scenarios*: During the 'idea generation' step, many good alternatives are never even considered at some companies. Many companies also fail to consider differing states of the world, which should ideally be incorporated through breakeven, scenario, and simulation analyses.

5. Real Options

Real options are options that allow managers to make decisions in the future that change the value of capital investment decisions made today. As with financial options, real options are contingent on future events. The difference is that real options deal with real assets.

Types of real options include:

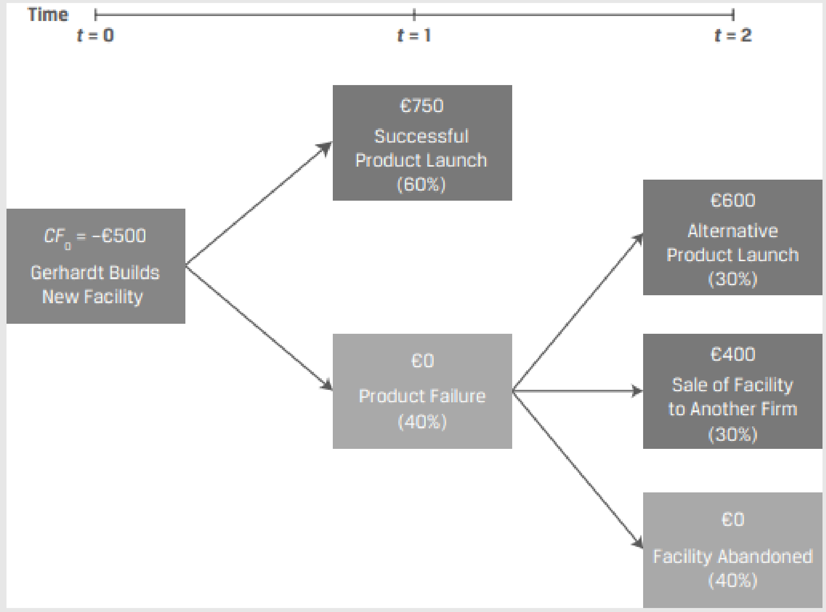
- Timing options: A company can delay investing until it has better information.
- Sizing options: If a company can invest in a project and then abandon it if its financial results are weak, it has an abandonment option. Conversely, if the company can make additional investments when financial results are strong, it has a growth option.
- Flexibility options: Once an investment is made, operational flexibilities such as changing the price (price setting option), or increasing production (production flexibility option) may be available.
- Fundamental options: In this case, the whole investment is an option. For example, the value of an oil well or refinery depends on the price of oil. If oil prices are low, a company may not drill a well. If oil prices are high, the company may pursue drilling.

There are several approaches to evaluating capital allocation projects with real options.

1. Use DCF analysis without considering options. If the NPV of the project without considering options is positive, then we can go ahead and make the investment. The presence of real options will simply add even more value. Therefore, it is not necessary to determine the value of the options separately.
2. If NPV is negative without considering options, then calculate project NPV as: Project NPV = NPV (based on DCF alone) – Cost of options + Value of options. Check if the project NPV turns positive after the options are considered.
3. Use decision trees and option pricing models. They can help in many sequential decision-making problems.

Example: Capital Allocation Using a Decision Tree
(*This is Example 9 from the curriculum.*)

Assume that Gerhardt Corporation is considering a €500 million outlay for a capital investment in a facility to produce a new product. Gerhardt assigns a 60% probability to a successful product launch, which is expected to return €750 million in one year's time. Gerhardt's finance team has also conducted an analysis of alternative facility uses, summarizing the timing and probability of cash flows associated with each real option in the following decision tree.



1. Calculate the NPV of Gerhardt's project without real options using a 10% required rate of return (r).
2. Calculate the NPV of Gerhardt's project with real options using a 10% required rate of return (r).

Solution to 1:

The NPV without real options is a probability-weighted cash flow if the product is successfully launched (60%) and a 40% probability that future cash inflows are zero.

$$\text{NPV} = \text{CF}_0 + \frac{\text{CF}_1}{(1+\text{r})^1} = -500 + \frac{(0.6 \times 750)}{1.10} = -90.91$$

Because NPV = –€90.91, Gerhardt should not pursue the project, based on the NPV decision rule.

Solution to 2:

$$\text{NPV} = \text{CF}_0 + \frac{\text{CF}_1}{(1+\text{r})^1} + \frac{\text{CF}_2}{(1+\text{r})^2}$$

$$\text{NPV} = -500 + \frac{(0.6 \times 750)}{1.10} + \frac{[(0.4 \times 0.3) \times 600] + [(0.4 \times 0.3) \times 400]}{(1.10)^2} = 8.26$$

The NPV with real options equals €8.26, which implies based on the NPV decision rule that Gerhardt should invest in the new production facility if alternative uses in the future are considered.

Summary

LO: Describe types of capital investments.

Companies invest for two primary reasons – to maintain their existing business and to grow it.

Projects undertaken by companies to maintain the business include:

- going concern projects
- regulatory/compliance projects

Projects undertaken by companies to expand the business include:

- expansion projects
- other projects

LO: Describe the capital allocation process, calculate net present value (NPV), internal rate of return (IRR), and return on invested capital (ROIC), and contrast their use in capital allocation.

Capital allocation is the process used by an issuer's management to make capital investment decisions.

The typical steps companies take in the capital allocation process are:

1. idea generation
2. investment analysis,
3. capital allocation planning
4. post-audit/monitoring

Net present value (NPV) is the present value of the future after-tax cash flows, minus the investment outlay (cost of the project). For independent projects, accept all projects with positive NPV. For mutually exclusive projects, accept the project with the higher NPV.

Internal rate of return (IRR) is the discount rate which makes NPV equal to 0. For independent projects, if IRR is greater than opportunity cost (required rate of return), accept the project, otherwise reject the project. For mutually exclusive projects, accept the project with the higher IRR as long as the IRR is greater than the opportunity cost.

ROIC reflects how effectively a company's management is able to convert capital into after-tax operating profits.

$$\text{ROIC} = \frac{\text{After tax operating profit}}{\text{Average invested capital}}$$

LO: Describe principles of capital allocation and common capital allocation pitfalls.

The key principles of capital allocation are:

- Decisions are based on after-tax cash flows.

- Measure incremental cash flows. Exclude sunk costs and include externalities.
- Timing of cash flows is crucial.

Common capital allocation pitfalls can be divided into cognitive errors and behavioral biases.

Cognitive errors

- Internal forecasting errors
- Ignoring cost of internal financing
- Inconsistent treatment of, or ignoring inflation

Behavioral biases

- Inertia
- Basing investment decisions on EPS, net income, or return on equity
- Pushing pet projects
- Failing to consider investment alternatives or alternative scenarios

LO: Describe types of real options relevant to capital investments.

The types of real options include:

1. Timing options
2. Sizing options - abandonment options or growth options
3. Flexibility options - price-setting options or production-flexibility options
4. Fundamental options

If NPV is positive without considering options, go ahead and make the investment.

If NPV is negative without considering options, calculate NPV (based on DCF alone) – Cost of options + Value of options.

LM06 Capital Structure

Ver 1.0

1. Introduction

This learning module covers:

- Calculating and interpreting the weighted-average cost of capital (WACC) for a company
- Factors affecting capital structure
- Modigliani–Miller propositions regarding capital structure
- Optimal and target capital structure

2. The Cost of Capital

Cost of capital is the rate of return that the suppliers of capital require as compensation for their contribution of capital. Assume a company decides to build a steel plant and needs money or capital for it. Investors such as bondholders or equity holders will lend this capital to the company. Suppliers of capital will be motivated to part with their money for a certain period of time if the money invested can earn a greater return than it would earn elsewhere. In short, investors will invest if the return (IRR) is greater than the cost of capital.

A company has access to several sources of capital such as issuing equity, debt, or instruments that share characteristics of both debt and equity. Each source becomes a component of the company's funding and has a specific cost associated with it called the **component cost of capital.**

The cost of capital is the rate of return expected by investors for average-risk investment in a company. Investors will demand a higher rate of return for higher-than-average-risk investments. Similarly, investors will demand a lower rate of return for lower-than-average-risk investments.

One way of calculating this cost is to determine the **weighted average cost of capital (WACC),** which is also called the **marginal cost of capital**. It is called marginal because it is the additional or incremental cost a company incurs to issue additional debt or equity.

Three common sources of capital are common shares, preferred shares, and debt. WACC is the cost of each component of capital in the proportion they are used in the company.

$$\text{WACC} = w_d r_d (1 - t) + w_p r_p + w_e r_e$$

where:

w_d = proportion of debt that the company uses when it raises new funds
r_d = before-tax marginal cost of debt
t = company's marginal tax rate
w_p = proportion of preferred stock the company uses when it raises new funds
r_p = marginal cost of preferred stock
w_e = proportion of equity that the company uses when it raises new funds
r_e = the marginal cost of equity

The weights are the proportions of the various sources of capital that the company uses. The weights should represent the company's **target capital structure** and not the current capital structure. For example, suppose that current capital structure of a company is 33.3% debt, 33.3% preferred stock and 33.3% common stock. To fund a new project, the company plans to issue more debt and its capital structure will change to 50% debt, 25% preferred stock, and 25% common stock. WACC calculations should be based on these new weights, i.e., the target weights.

Example

IFT has the following capital structure: 30 percent debt, 10 percent preferred stock, and 60 percent equity. IFT wants to maintain these weights as it raises additional capital. Interest expense is tax deductible. The before-tax cost of debt is 8 percent, cost of preferred stock is 10 percent, and cost of equity is 15 percent. If the marginal tax rate is 40 percent, what is the WACC?

Solution:

WACC = (0.3) (0.08) (1 - 0.4) + (0.1) (0.1) + (0.6) (0.15) = 11.44 percent

Note: Before-tax cost of debt is given. Do not forget to calculate the after-tax cost.

Example

Machiavelli Co. has an after-tax cost of debt capital of 4 percent, a cost of preferred stock of 8 percent, a cost of equity capital of 10 percent, and a weighted average cost of capital of 7 percent. Machiavelli Co. intends to maintain its current capital structure as it raises additional capital. In making its capital budgeting decisions for the average risk project, what is the relevant cost of capital?

Solution:

The relevant cost of capital is 7%. The WACC using weights derived from the current capital structure is the best estimate of the cost of capital for the average risk project of a company.

3. Factors Affecting Capital Structure

Capital structure refers to the specific mix of debt and equity a company uses to finance its assets and operations. Issuers desire a capital structure that minimizes their weighted-average cost of capital and generally matches the duration of their assets.

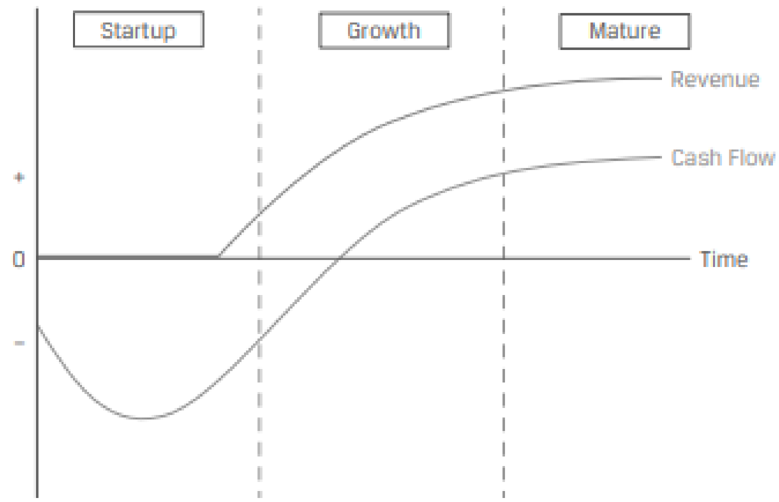
Determinants of the Amount and Type of Financing Needed

The total amount and type of financing needed usually depends on the issuer's business model and its position in the corporate life cycle.

Corporate Life Cycle

A company's life cycle stage influences its cash flow characteristics, its ability to support debt; and is therefore a primary factor in determining capital structure. Any capital that is not sourced through borrowing must come from equity.

As companies mature and move from start-up, through growth, to maturity, their business risk typically declines, and their operating cash flows turn positive and become more predictable. This allows for greater use of leverage at more attractive terms. This is illustrated in Exhibit 2 from the curriculum.



Feature / Phase	Startup	Growth	Mature
Revenue Growth	Initial	Rising	Slowing
Free Cash Flow	Negative	Rising	Stable
Business Risk	High	Medium	Low
Debt Availability	Little or None	Rising	High
Debt Type	Convertible	Secured	Unsecured

Start-Ups

- In this stage a company's revenues are close to zero and a lot of investment is required to move from the prototype stage to commercial production.
- Therefore, cash flow is usually negative.
- The risk of business failure is high.
- The company typically raises capital through equity rather than debt.
- Equity is generally sourced through private markets (venture capital) rather than public markets.
- Debt is generally not available or is very expensive. It is usually a negligible component of the capital structure.
- Some start-ups may be able to raise convertible debt.

<u>Growth Businesses</u>

- In this stage a company typically experiences high revenue growth but investment is needed to achieve this growth.
- Therefore, cash flow may be negative but it is likely to be improving and becoming more predictable.
- The risk of business failure decreases.
- As the business becomes more attractive to lenders, debt financing may be available at reasonable terms. The company may also have assets that it can use to secure debt.
- However, most companies use debt conservatively to retain their operational and financial flexibility.
- Equity is generally the main source of capital.

<u>Mature Businesses</u>

- In this stage a company typically experiences a slowdown in revenue growth; and growth-related investment spending decreases.
- Cash flow is usually positive and predictable.
- The risk of business failure is low.
- Debt financing is available at attractive terms often on an unsecured basis.
- To take advantage of the cheaper debt (as compared to equity) companies typically use significant leverage.
- Over time a company may experience de-leveraging due to continuous positive cash flow generation and share price appreciation.
- To offset this de-leveraging companies typically buy back shares and reduce the proportion of equity in the capital structure.
- Share buy backs are preferred over cash dividends as they are more tax-efficient and do not set future expectations.
- Investors generally respond favorably to share buyback announcement, which may lead to an increase in share price.

In the section above we established a general relationship between company maturity and capital structure. However, there are two important exceptions:

Capital-Intensive Businesses

Business such as real estate, utilities, shipping, airline etc. are highly capital intensive. Also, the underlying assets can be bought and sold fairly easily and make for a good collateral. Such businesses tend to use high levels of leverage irrespective of their maturity stage.

Capital-Light Businesses

Some businesses such as software can scale easily and do not require substantial investments in fixed or working capital to support growth. They are typically cash flow positive from an early stage and never need to raise large amounts of capital. Therefore, they tend to use very little debt in their capital structure.

Determinants of the Costs of Debt and Equity

The costs of debt and equity are determined in the financial markets by top-down factors that affect the overall market, as well as investor's assessment of issuer-specific factors.

Top-Down Factors

A company's cost of debt is equal to a benchmark risk-free rate plus a credit spread specific to the company. Market conditions/ business cycle affect both the benchmark rates and the credit spreads. The credit spreads tend to widen during recessions and tighten during expansions.

Companies may increase their use of debt when borrowing is less expensive due to low benchmark interest rates and/or tight credit spreads, and vice versa.

Issuer-Specific Factors

Investors consider the risk and return profile of an issuer and adjust their required rates of return relative to a base rate by evaluating the following factors:

- <u>Sales risk</u>: Some companies like Vodafone in the telecom industry, have very stable revenue streams because a large proportion of their revenues are subscription-like, recurring revenues. This is generally viewed as a positive. In contrast, companies in cyclical industries, such as Toyota in the automobile industry, have more volatile revenue streams that are highly sensitive to fluctuations in the business cycle. This is generally viewed as a negative.
- <u>Profitability risk</u>: Aside from revenue stability, profit margin stability is an important factor, which is determined by a company's proportion of fixed versus variable costs. Operating leverage = Fixed costs/Total costs. Companies with higher operating leverage experience a greater change in cash flow and profitability for a given change in revenue than firms with low operating leverage
- <u>Financial leverage and interest coverage</u>: The existing financial leverage of a firm influences its capital structure decisions. Highly leveraged firms face a higher risk of default and, as a result, have less capacity to service additional debt. Underleveraged firms, on the other hand, can support additional debt relatively easily.
- <u>Collateral/type of assets owned by the firm</u>: In general, assets that support increased debt use include those that are considered strong collateral, generate cash, or are fungible or liquid, such as real estate, automobiles, aircraft, and receivables from creditworthy customers.

4. Modigliani–Miller Capital Structure Propositions

Economists Modigliani and Miller claimed that given certain assumptions, a company's choice of capital structure does not affect its value.

The assumptions made were:

1. Homogeneous expectations: All investors have the same expectations with respect to the cash flows from an investment in bonds or stock.
2. Perfect capital markets: There are no transaction costs, no taxes, no bankruptcy costs, and everyone has the same information.
3. Risk-free rate: Investors can borrow and lend at the risk-free rate.
4. No agency costs: Managers always act to maximize shareholder wealth.
5. Independent decisions: Financing and investment decisions are independent of each other. Operating income is unaffected by changes in the capital structure.

Modigliani and Miller's work provides us a starting point and allows us to examine what happens when the assumptions are relaxed to reflect real-world considerations.

Capital Structure Irrelevance (MM Proposition I without Taxes)

MM Proposition I: The market value of the company is not affected by the capital structure of the company. This is because individual investors can create any capital structure they prefer for the company by borrowing and lending in their own accounts in addition to holding shares in the company.

$V_L = V_U$

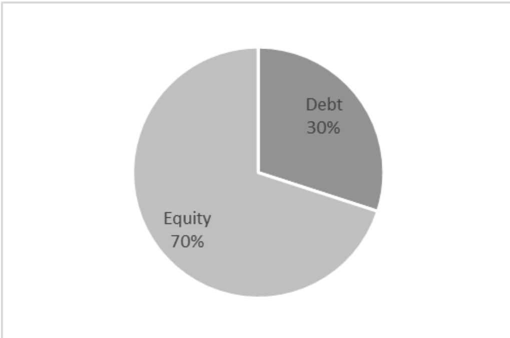
where:
V_L = value of the levered firm
V_u = value of the unlevered firm

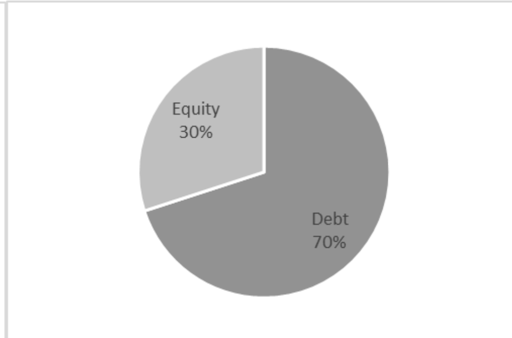
In other words, the value of a company is determined solely by its cash flows, not by the relative reliance on debt and equity capital.

The weighted average cost of capital (WACC) is unaffected by the capital structure.

where: $WACC = w_d\, r_d + w_p r_p + w_e r_e$

We can explain MM's capital structure irrelevance proposition in terms of a pie. Suppose the total value of a company is represented by a pie. Slices represents how much of the total capital is contributed by debt and equity. Depending on the capital structure, the pie can be split in any number of ways, but the size of the pie will remain the same. This is illustrated in the figures below.





Higher Financial Leverage Raises the Cost of Equity (MM Proposition II without Taxes)

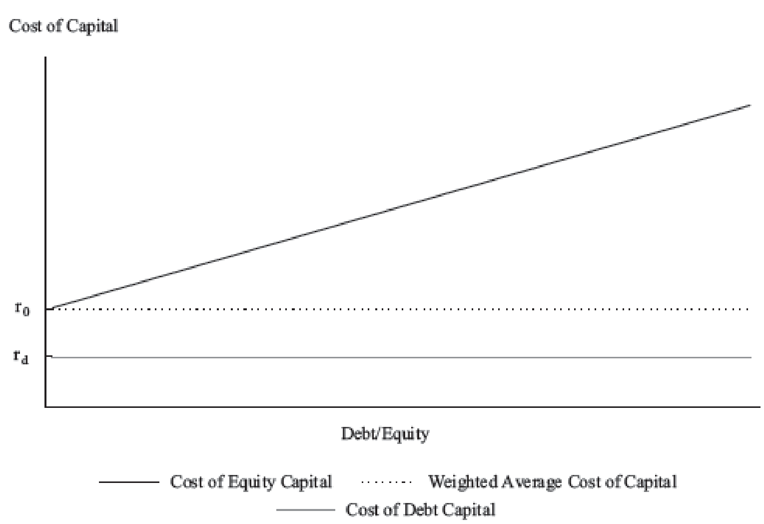
MM Proposition II: The cost of equity is a linear function of the company's debt-to-equity ratio.

$$r_e = r_0 + (r_0 - r_d)\frac{D}{E}$$

where:
r_0 is the cost of capital for a company financed only by equity and has zero debt
r_d is the cost of debt
r_e is the cost of equity
D/E = debt-to-equity ratio

As D/E rises, i.e. the company increases the use of debt, the cost of equity (r_e) rises. We know from MM Proposition I that the value of a company is unaffected by changes in D/E and the WACC remains constant. Proposition II then implies that the cost of equity increases in such a manner as to exactly offset the increased use of cheaper debt in order to maintain a constant WACC. Under this proposition, WACC is determined by the business risk of the company, and not by the capital structure.

MM Proposition II is illustrated in the figure below.



As leverage increases, the cost of equity increases, but WACC and the cost of debt remain constant.

Firm Value with Taxes (MM Proposition II with Taxes)

The discussion so far has ignored taxes. Now, we will present MM propositions I and II with taxes. As interest paid is tax deductible, the use of debt provides a tax shield that increases the value of a company. If we ignore the costs of financial distress and bankruptcy, the value of the company increases as we take on more debt.

MM Proposition I with taxes: The value of a levered company is equal to the value of an unlevered company plus the value of the debt tax shield.

$$V_L = V_U + tD$$

where:
V_L = value of the levered firm
V_u = value of the unlevered firm
t = marginal tax rate
D = value of debt in the capital structure
The term tD is often referred to as the debt tax shield

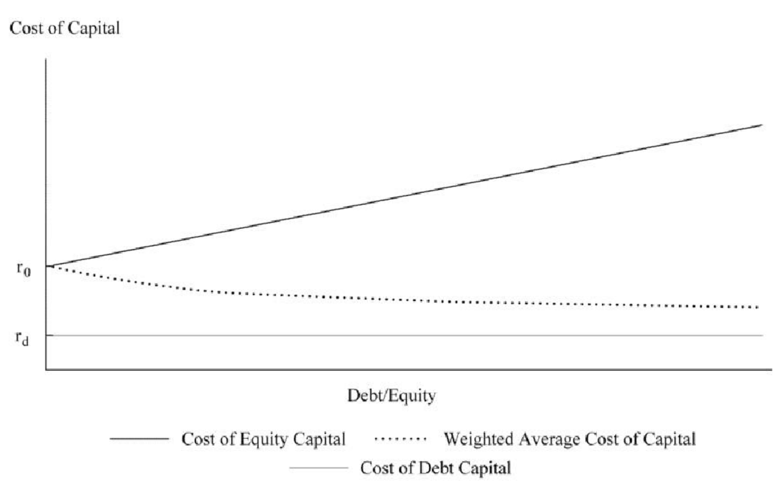
Cost of Capital (MM Proposition II with Taxes)

MM Proposition II with taxes: The cost of equity is a linear function of the company's debt-to-equity ratio with an adjustment for the tax rate.

The cost of equity increases as the company increases the amount of debt in its capital structure, but the cost of equity does not rise as fast as it does in the no tax case.

$$r_e = r_0 + (r_0 - r_d)(1 - t)\frac{D}{E}$$

WACC for a leveraged company falls as debt increases, and therefore the overall company value increases. This is illustrated in the figure below:



This proposition implies that in the presence of taxes (but no financial distress or bankruptcy costs), the use of debt is value enhancing and, at the extreme, 100% debt is optimal.

The table below provides a summary of MM propositions.

	Without Taxes	**With Taxes**
Proposition I	$V_L = V_U$	$V_L = V_U + tD$
Proposition II	$r_e = r_0 + (r_0 - r_d)\ D/E$	$r_e = r_0 + (r_0 - r_d)(1 - t)\ D/E$

Cost of Financial Distress

The disadvantage of operating and financial leverage is that earnings are magnified downward during economic slowdown. Lower or negative earnings put companies under stress, and this financial distress adds costs to a company.

The costs of financial distress can be classified into direct and indirect costs. Direct costs include the actual cash expenses associated with the bankruptcy process, such as legal and administrative fees. Indirect costs include forgone investment opportunities, impaired ability to conduct business, and agency costs associated with the debt during periods in which the company is near or in bankruptcy.

The costs of financial distress are lower for companies whose assets have a ready secondary market. For example, airlines, shipping companies etc.

The probability of financial distress and bankruptcy increases as the degree of leverage increases. The probability of bankruptcy depends, in part, on the company's business risk. Other factors that affect the likelihood of bankruptcy include the company's corporate governance structure and the management of the company.

5. Optimal Capital Structure

So far in our discussion of the MM propositions, we have only relaxed the assumption of no corporate taxes. We now consider a scenario that includes both corporate taxes and the costs of bankruptcy/financial distress.

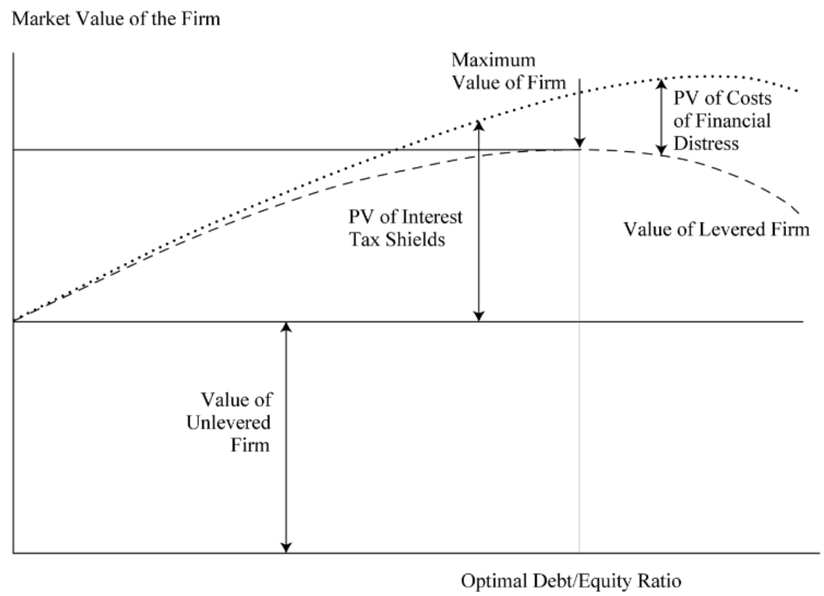
The optimal capital structure is the one at which the value of the company is maximized. The static trade-off theory is based on balancing the expected costs from financial distress against the tax benefits of debt service payments. Considering both the tax shield provided by debt and the costs of financial distress, the expression for the value of a leveraged company becomes:

$$V_L = V_U + tD - PV(\text{Costs of financial distress})$$

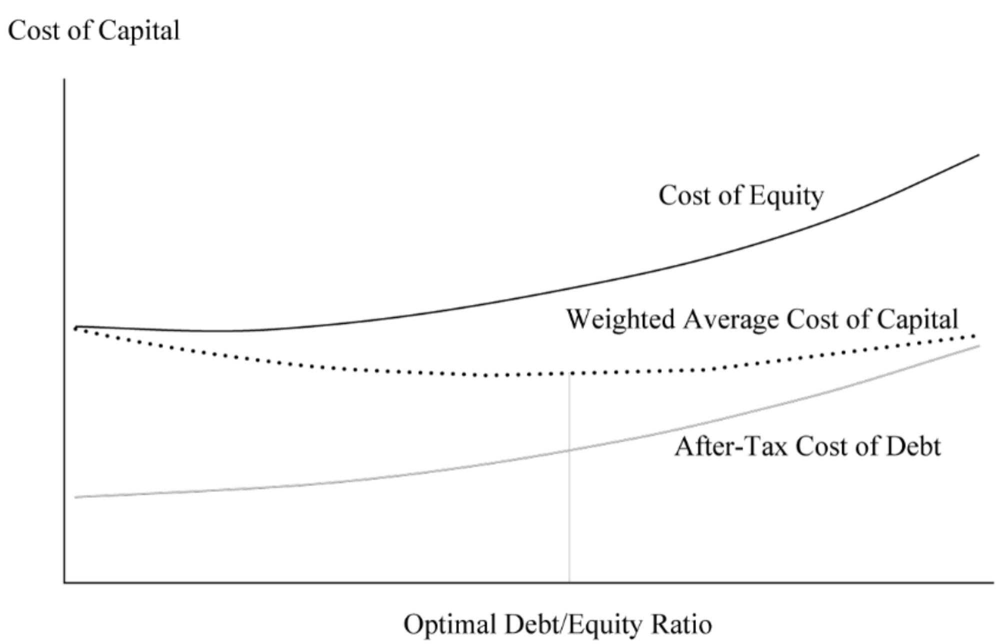
This equation represents the static trade-off theory of capital structure. It results in an optimal capital structure where debt is less than 100%. At low levels of debt, the tax benefits of debt outweigh the PV of financial distress, and the firm's value increases.

However, as more debt is added, the PV of financial distress begins to rise. At the optimal debt/equity ratio point, the PV of financial distress is equal to the tax benefit of debt, and the firm value is maximum. If more debt is added beyond this point, the PV of financial distress outweigh the tax benefits of debt, and the firm's value decreases.

These concepts are illustrated in Exhibit 10 from the curriculum.



WACC is minimum at the point where the firm value is maximized.



When a company identifies its most appropriate capital structure, it may adopt this as its target capital structure. However, a company's capital structure may vary from its target because management may try to take advantage of short-term opportunities in alternate financing sources. Market value variations also continuously affect the company's capital structure. Sometimes, it may be impractical and expensive for a company to maintain its target structure.

In practice, it is difficult to precisely determine the optimal capital structure, because of the difficulty in estimating some costs, such as the costs of financial distress. Therefore, managers often use an optical capital structure range instead of a precise ratio. For example, instead of saying exactly 40% debt is optimal, managers can say debt should be in the range of 30% to 50%.

Market Value vs. Book Value

The optimal capital structure should be calculated using the market value of equity and debt. However, company capital structure targets often use book values instead because:

- Market values can fluctuate a lot and they do not necessarily impact the appropriate level of borrowing. For example, a company that has seen a rapid share price increase may decide to take advantage of this situation and issue even more equity instead of debt.
- The management is primarily concerned with the amount and types of capital invested 'by' the company and not 'in' the company. Their perspective is different from investors who have purchased securities at the prevailing market price.
- Lenders and rating agencies typically focus on the book value of debt and equity for their calculation measures.

Financing decisions are often opportunistic. Managers consider the share price of their company as well as market interest rates for their debt when deciding when, how much, and what type of capital to raise.

Target Weights and WACC

When conducting an analysis if we know the company's target capital structure, then we should use it in our analysis. However, analysts typically do not know a company's target capital structure. It can be estimated using one of these methods:

1. Assume the company's current capital structure, at market value weights for the components, represents the company's target capital structure.
2. Examine trends in the company's capital structure or statements by management regarding capital structure policy to infer the target capital structure.
3. Use averages of comparable companies' capital structures as the target capital structure.

Example: Estimating the Proportions of Capital

(*This is based on knowledge check example from the curriculum.*)

The following information is provided for a company:

Market value of debt: EUR50 million

Market value of equity: EUR60 million

Primary competitors and their capital structures (in millions):

Competitor	Market Value of Debt	Market Value of Equity
A	EUR25	EUR50
B	EUR101	EUR190
C	GBP40	GBP60

Calculate the proportions of debt and equity, if the target capital structure is calculated based on:

1. The current capital structure

2. Competitor's capital structure

3. The company's announcement that a debt-to-equity ratio of 0.7 reflects its target capital structure.

Solution to 1:

$$w_d = \frac{50}{50 + 60} = 0.4545$$

$$w_e = \frac{60}{50 + 60} = 0.5454$$

Solution to 2:

$$w_d = \frac{\left(\frac{25}{25+50}\right) + \left(\frac{101}{101+190}\right) + \left(\frac{40}{40+60}\right)}{3} = 0.3601$$

$$w_e = \frac{\left(\frac{50}{25+50}\right) + \left(\frac{190}{101+190}\right) + \left(\frac{60}{40+60}\right)}{3} = 0.6399$$

Solution to 3:

A debt-to-equity ratio of 0.7 represents a weight on debt of 0.7/1.7 = 0.4118, so w_d = 0.4118 and w_e = 1 − 0.4118 = 0.5882.

Pecking Order Theory and Agency Costs

Managers have more information about a company's performance and prospects than outsiders, such as owners and investors. This is referred to as information asymmetry.

Investors demand higher returns when asymmetry of information is high because this increases the probability of agency costs.

Investors closely watch manager's decision for insights into the company's future prospects. Managers may provide information to investors ("signaling") through their choice of financing method. Fixed payment commitments, for example, can indicate management's confidence in the company's future prospects.

Being aware of this, managers take into account how their actions might be interpreted by outsiders.

Pecking Order Theory: This theory suggests that managers choose methods of financing that send the least signal to outsiders. The preferred hierarchy for financing is:

- First preference: internal financing (retained earnings)
- Second preference: debt financing
- Third preference: equity financing

Agency Costs: Agency costs are incremental costs that arise from conflicts of interest between managers, shareholders, and bondholders. Agency theory predicts that as the use of debt increases, agency costs to equity will decrease. The more financially leveraged a company is, the less freedom managers have to incur additional debt or waste cash in inefficient ways. This also serves as the basis for the 'free cash flow hypothesis.'

Free Cash Flow Hypothesis: As per this hypothesis, high debt levels discipline managers by forcing them to manage the company efficiently so that the company can make its interest and principal payments. Thus, by reducing the company's free cash flows, managers have fewer opportunities to misuse cash.

Summary

LO: Calculate and interpret the weighted-average cost of capital for a company.
An issuer's cost of capital is estimated using a weighted average of the costs of debt and equity, using either the current market value or management's target weights of each type of financing as the weights.

$\text{WACC} = w_d r_d (1 - t) + w_p r_p + w_e r_e$

WACC represents the overall cost of capital for the firm and is the appropriate discount rate to use for projects having a similar risk profile as that of the firm.

LO: Explain factors affecting capital structure and the weighted-average cost of capital.

Capital structure refers to the specific mix of debt and equity a company uses to finance its assets and operations. Issuers desire a capital structure that minimizes their weighted-average cost of capital and generally matches the duration of their assets.

The total amount and type of financing needed usually depends on the issuer's business model and its position in the corporate life cycle. As a company matures and progresses from start-up to growth to maturity, its business risk typically decreases, and its operating cash flows turn positive and more predictable. This enables greater use of leverage at more favorable terms. Capital intensive businesses tend to use high levels of leverage irrespective of their maturity stage. Whereas, capital light businesses tend to use very little debt in their capital structure.

The component cost of debt and equity are impacted by top-down factors such as financial market and industry conditions; and by issuer specific factors such as stability of revenues and profit margins, and operating and financial leverage.
LO: Explain the Modigliani–Miller propositions regarding capital structure.

	Without Taxes	With Taxes
Proposition I	$V_L = V_U$	$V_L = V_U + tD$
Proposition II	$r_e = r_0 + (r_0 - r_d)\, D/E$	$r_e = r_0 + (r_0 - r_d)(1 - t)\, D/E$

LO: Describe optimal and target capital structures.

The optimal capital structure is the one at which the value of the company is maximized. The static trade-off theory is based on balancing the expected costs from financial distress against the tax benefits of debt service payments. Considering both the tax shield provided by debt and the costs of financial distress, the expression for the value of a leveraged company becomes:

$$V_L = V_U + tD - \text{PV(Costs of financial distress)}$$

At the optimal debt/equity ratio point, the PV of financial distress is equal to the tax benefit of debt, and the firm value is maximum.

LM07 Business Models

Version 1.0

1. Introduction

This reading covers:

- What is a business model
- The types of business models

A well-defined business model helps analysts in understanding a company's operations, strategy, target customers, key partners, prospects, risks, and financial profile. Many companies have conventional business models that are easily understood, such as manufacturer, wholesaler, retailer, restaurant chain etc.

However, the advent of digital technology has changed the way most businesses operate, and enabled disruption of existing business models. Many companies now have business models that are complex, specialized, or new.

2. Defining the Business Model

A business model describes how a business is organized to deliver value to its customers:

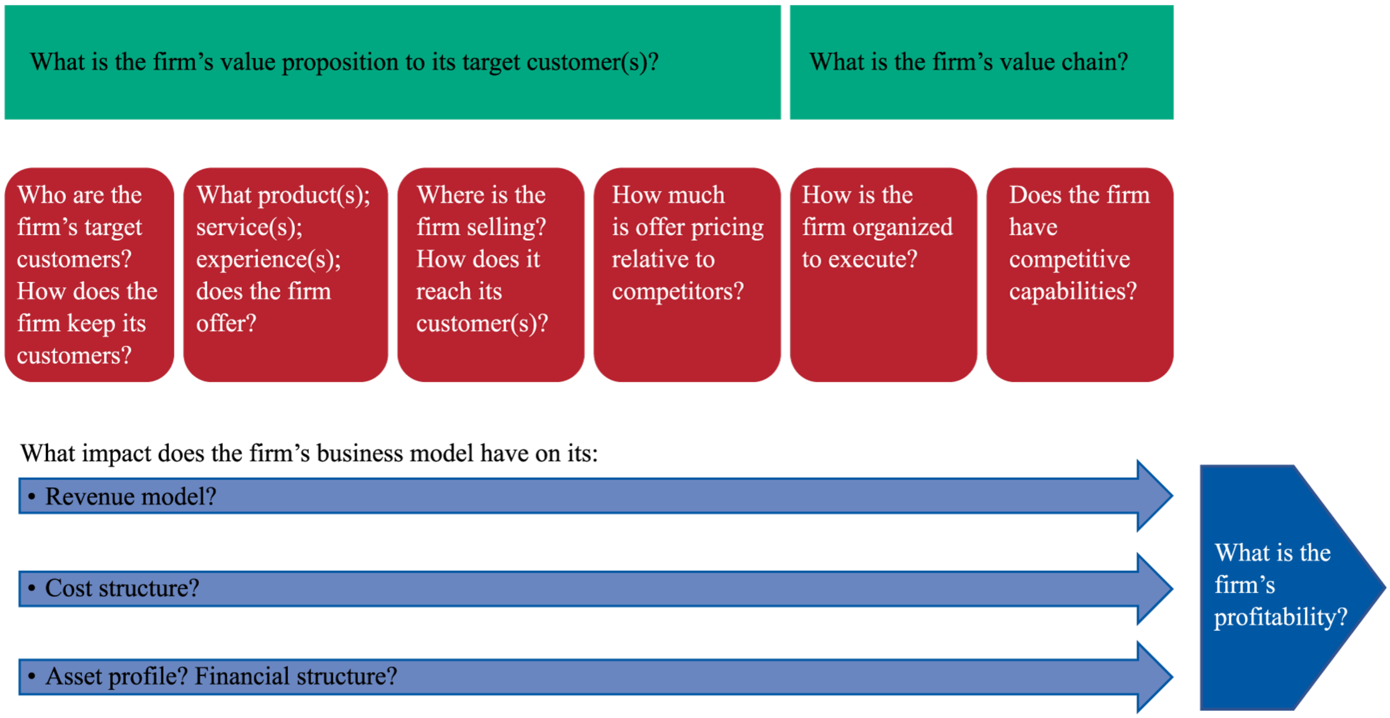
- who its customers are,
- how the business serves them,
- key assets and suppliers, and
- the supporting business logic.

A business model explains what the company does, how it operates, and how it generates revenue and profits, as well as how it differs from competitors. It provides enough detail to understand the basic relationships between these key elements, however, it does not provide a full description. For a full description (such as detailed financial forecasts) we would have to refer to a business plan.

A business model should have a value proposition and a value chain.

- The firm's "value proposition" refers to the product or service attributes valued by a firm's target customer that lead those customers to prefer a firm's offering over those of its competitors, given relative pricing.
- The firm's "value chain" refers to how the firm is structured to deliver that value. It refers to the systems and processes within a firm that create value for its customers.

This is illustrated in Exhibit 1 from the curriculum.



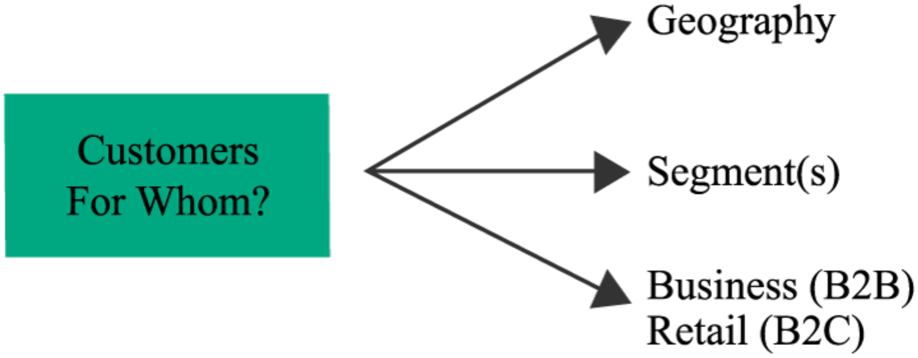
Business Model Features

The key features of a public company's business model are often provided in annual reports or other disclosure documents. As an example, the curriculum presents the 'business description' from Tesla's annual report.

> "We design, develop, manufacture, sell and lease high-performance fully electric vehicles and energy generation and storage systems, and offer services related to our products. We are the world's first vertically integrated sustainable energy company, offering end-to-end clean energy products, including generation, storage and consumption. We generally sell our products directly to customers, including through our website and retail locations. We also continue to grow our customer-facing infrastructure through a global network of vehicle service centers, Mobile Service technicians, body shops, Supercharger stations and Destination Chargers to accelerate the widespread adoption of our products. We emphasize performance, attractive styling and the safety of our users and workforce in the design and manufacture of our products, and are continuing to develop full self-driving technology for improved safety. We also strive to lower the cost of ownership for our customers through continuous efforts to reduce manufacturing costs and by offering financial services tailored to our vehicles. Our sustainable energy products, engineering expertise, intense focus to accelerate the world's transition to sustainable energy and achieve the benefits of autonomous driving, and business model differentiate us from other companies."

In this paragraph, Tesla describes many features, including its products and their attributes, key channels, and its emphasis on innovation and vertical integration. Let's take a closer look at the features.

Customers, Market: Who

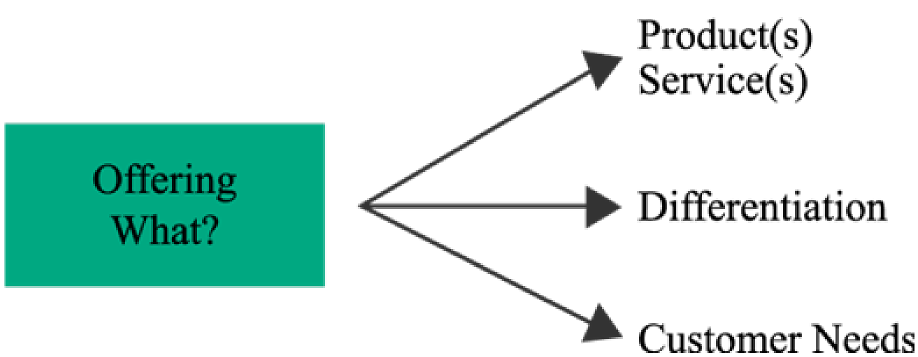


A business model should identify the firm's target customers:

- What geographies will be served?
- What market segments will be served? (e.g., high income suburban families)
- What customer segments will be served? Is this a business (B2B) or consumer (B2C) market?

In the Tesla example, the business description does not specify which customer segments are being targeted. This is most likely because Tesla's target market is shifting over time toward the mass market, as costs and prices decline.

Firm Offering: What

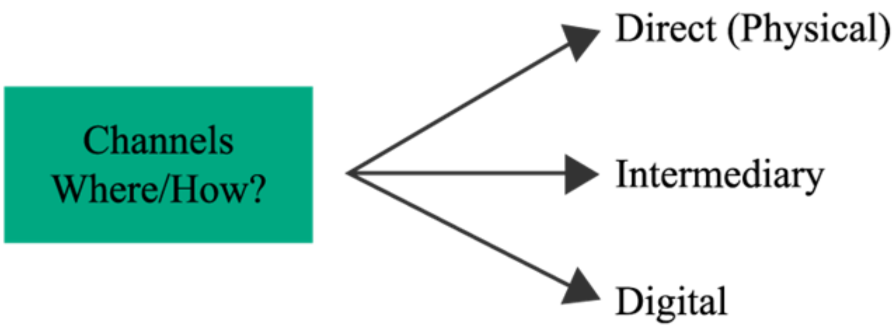


A business model should define:

- What products and services are being offered?
- What differentiates these offerings from competitors?
- How are the needs of its target customers being met?

In the Tesla example, instead of using a broad description such as "electric car" to describe its product offering; Tesla has used a more precise description that is useful to analysts - "high-performance fully electric vehicles and energy generation and storage systems".

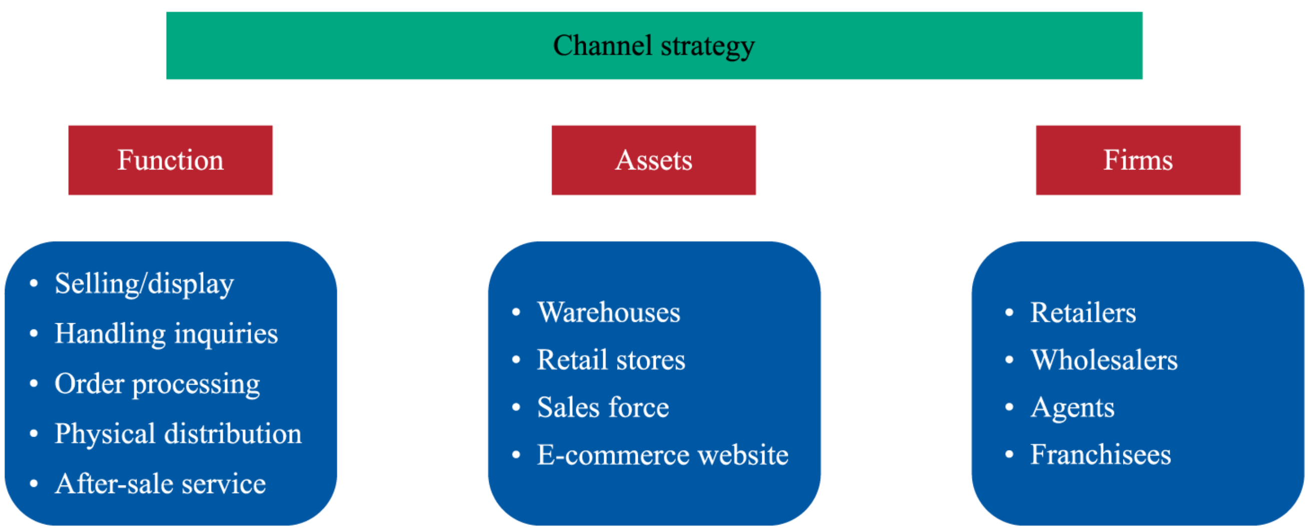
Channels: Where



A firm's channel strategy refers to "where" the firm is selling its offering and how it is reaching its customers. Channel strategy typically involves two main functions:

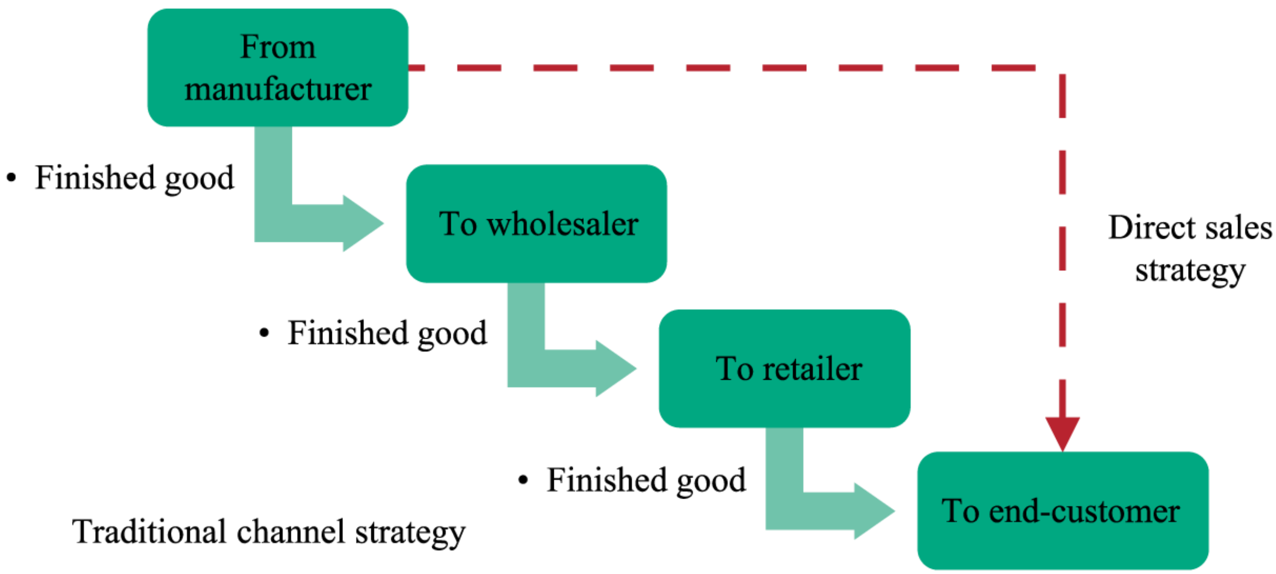
- Selling the firm's products and services
- Delivering them to customers

When evaluating a firm's channel strategy, it is important to distinguish the functions performed, from the assets that might be involved, and different firms that might be involved in performing those functions or owning those facilities. Exhibit 2 from the curriculum, illustrates this concept.



Many "product" businesses, employ **a traditional channel strategy**, that reflects the flow of finished goods (e.g., from manufacturer to wholesaler, retailer, and end customer). However, some manufacturers may employ a **direct sales** strategy, selling directly to the end customer. This strategy bypasses (or disintermediates) the distributor and retailer.

Typically, direct sales involved the use of the company's own sales force and was very expensive. However, with e-commerce, direct sales have now become a cost-effective strategy. Exhibit 3 from the curriculum compares the two strategies.



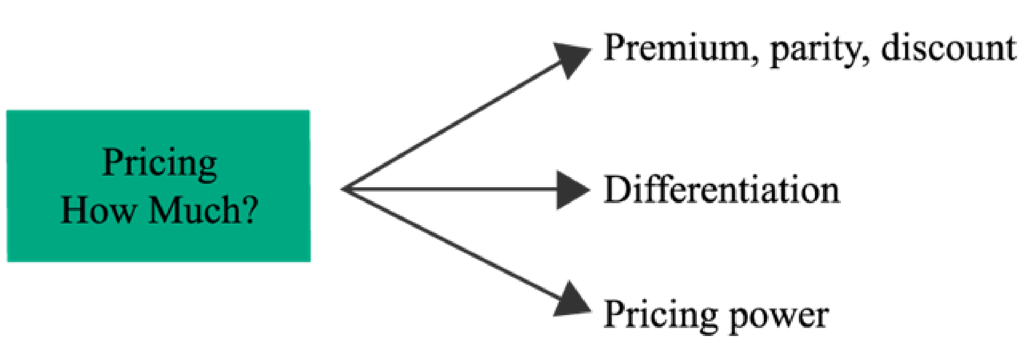
When an intermediary is involved, that intermediary may work on an **agency basis**, earning commissions, rather than taking ownership of the goods (e.g., auctioneers such as Sotheby's that deal in fine art).

The **drop shipping** model in e-commerce allows an online marketer to have goods delivered directly from the manufacturer to the end customer without taking the goods into inventory.

Companies often use several channels in tandem. With an **omnichannel** strategy, both digital and physical channels are used to complete a sale. For example, a customer might order an item online and pick it up in a store ("click and collect").

It is important to understand how a firm's channel strategy differs from those of its competitors. For example, Tesla mentions its direct sales strategy, which differs from the franchised dealer model used by most automakers.

Pricing: How Much



A business model should answer the following questions related to pricing:

- Does the firm price at a premium, parity, or discount relative to competitors?
- How is the firm's pricing justified in its business model?

Based on pricing power companies can be classified into:

- **Price takers –** Companies with little differentiation are "commodity" producers who must accept market prices dictated to them. Such companies focus on other sources of value such as employing a discounting strategy to build scale and a cost advantage (e.g., Walmart).
- **Price setters –** Companies with high differentiation can command premium pricing and face significantly less pricing risk from competitors.

Most firms try to differentiate their offerings in some way to achieve some degree of pricing power. For example, Tesla focuses on the **"total cost of ownership"** which factors in government subsidies and lower operating costs for electric vehicles as an offset to higher purchase prices.

Pricing and Revenue Models

Pricing approaches are typically value based or cost based.

- *Value-based pricing* attempts to set pricing based on the value received by the customer. For example, Tesla can command a premium price because they have low operating costs, which is a source of value for their customers.
- *Cost-based pricing* attempts to set pricing based on costs incurred. For example, an environmental law firm may charge clients by the hour because it is difficult to predict how many hours (cost) will be required and how much value the client will receive.

Price Discrimination

Price discrimination refers to situations where firms charge different prices to different customers. The objective of price discrimination is to maximize revenues in a situation where different customers have different willingness to pay. Common pricing strategies in this category include:

- *Tiered pricing* charges different prices to different buyers, most commonly based on purchase volume.
- *Dynamic pricing* charges different prices at different times. For example, off-peak pricing offered by hotel rooms and airlines, and movie tickets; and surge pricing offered by ride sharing companies.
- *Auction/reverse auction models* establish prices through bidding.

Pricing for Multiple Products

Firms selling multiple products frequently use the following pricing models:

- *Bundling* refers to combining multiple products or services so that customers are incentivized, or required, to buy them together. For example, cable TV and internet services, prepackaged set of kitchen utensils.
- *Razors-and-blades pricing* combines a low price on a piece of equipment and high-margin pricing on repeat-purchase consumables. For example, razors and blades, printers and refill cartridges.
- *Optional product pricing* applies when a customer buys additional services or product features, either at the time of purchase or afterward. For example, deluxe interior for a car, a side order with a restaurant meal.

Pricing for Rapid Growth

- *Penetration pricing*: A firm willingly sacrifices margins in order to build scale and market share. For example, Netflix subscriptions, Amazon e-readers.
- *Freemium pricing*: Allows customers a certain level of usage or functionality at no charge. For example, mobile games, software applications.
- *Hidden revenue business models*: Provide free services to users while generating revenue elsewhere. For example, media sector which provides free content and paid advertising.

Alternatives to Ownership

Some business models create value by offering an alternative to purchasing an asset or product, such as:

- *Recurring revenue/subscription pricing*: Enables customers to rent a product or service for as long as they need it. For example, TV channels, telecommunication services.
- *Fractionalization*: Creates value by selling an asset in smaller units or through the use of an asset at different times. For example, web hosting which allows sharing of server capacity, Office sub-leasing/co-working.
- *Leasing*: Involves shifting the ownership of an asset from the firm using it to an entity that has lower costs for capital and maintenance. For example, real estate, aircrafts.
- *Licensing*: Gives a firm access to intangible assets (e.g., brand name, song, patented formula) in return for royalty payments.

- *Franchising*: It is a more comprehensive form of licensing, in which the franchisor typically gives the franchisee the right to sell or distribute its product or service in a specified territory and to receive marketing and other support.

Value Proposition (Who + What + Where + How Much)

A firm's value proposition refers to the product or service attributes valued by a firm's target customer that lead those customers to prefer a firm's offering over those of its competitors, given relative pricing.

Value propositions can arise from:

- The product itself – e.g., performance, features style etc.
- Service and support – e.g., access to repairs, spare parts etc.
- The sale process – e.g., purchasing convenience, no-hassle return
- Pricing relative to competitors

In the Tesla example, its electric car value proposition highlights the advantages of its electric propulsion system: zero emissions and high performance (strong and silent acceleration) and technological sophistication (e.g., self-driving capabilities, frequent enhancements via software upgrades).

Business Organization, Capabilities: How

In addition to specifying a firm's value proposition, a business model should also specify "how" the firm is structured to deliver that value. It should address the following questions:

- What assets and capabilities (e.g., skilled personnel, technologies) does the firm require to execute on its business model?
- Will these be owned/insourced or rented/outsourced?

Value Chain

The "how" aspect of a business model is also referred to as a firm's value chain. It refers to the systems and processes within a firm that create value for its customers.

A value chain includes only those functions performed by a single firm. Some of these functions may be valued by customers but may not involve physical transformation or handling the product.

A firm's value chain is different from a *supply chain*. A supply chain refers to the sequence of processes involved in the creation of a product, both within and external to a firm. It includes all steps involved in producing and delivering a product, regardless of whether those steps are performed by a single firm.

Value chain analysis provides a link between the firm's value proposition for customers and its profitability. It involves:

- identifying the specific activities carried out by the firm,

- estimating the value added and costs associated with each activity, and
- identifying opportunities for competitive advantage.

Profitability and Unit Economics

A business model should specify how the firm expects to generate its profits. To understand the profitability of a business, the analyst should examine margins, break-even points and unit economics (which is expressing revenues and costs on a per-unit basis).

The curriculum provides the following examples for unit economics:

Example:

A producer of bottle caps might sell its product at 2.5 cents per unit, with direct costs for material and labor of 2.0 cents per unit and a contribution margin (selling price per unit minus variable cost per unit) of 0.5 cents per unit. If the firm has fixed costs of USD500,000 per year, what is its unit break-even point?

Solution:

Break-even point (unit) = Fixed costs/Contribution margin

= USD500,000/0.5 cents

= 100 million units.

Example:

A restaurant chain might have an average order of EUR50, with ingredient costs equal to 50% of sales. If fixed costs are EUR250,000 annually per outlet, what is the firm's unit break-even point and operating margin at 20,000 orders per year?

Solution:

Break-even point (order) = Fixed costs/Contribution margin

= EUR250,000/EUR25

= 10,000 orders/year.

Operating margin = 20,000 orders × EUR50

= EUR1,000,000 revenues – EUR500,000 ingredient costs (50% of sales) – EUR250,000 fixed costs

= EUR250,000 operating profit, or

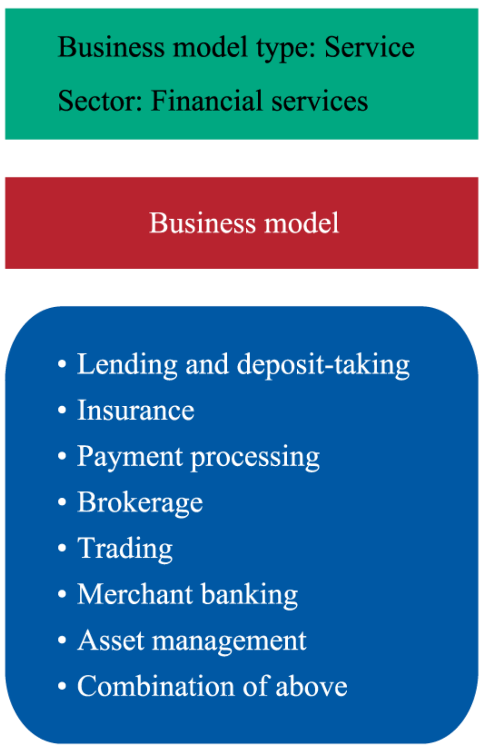
= EUR250,000/EUR1,000,000 = 25%.

Tesla's business model is based on declining unit revenues and costs over time as volume increases and technology improves. This is expected to result in a virtuous circle: Lower prices allow Tesla to expand its addressable market and market share, while lower costs allow profits to rise and create a competitive barrier.

3. Business Model Types

Each industry tends to has its own set of established business models. Firms in the goods-producing sectors are generally easy to classify based on how they fit into the supply chain. For example, manufactures, wholesalers, retailers, suppliers etc.

However, service businesses are more diverse. For example, the financial services sector has many established business models, as shown in the exhibit below.



Some firms combine these models together (e.g. universal banks) while some specialize in a particular model (e.g. discount brokers).

Business Model Innovation

Digital technology has transformed the "where" and "how" elements of business models in established markets, by drastically reducing the cost of communicating, exchanging information, and transacting between businesses.

- *Location matters less*: customers can easily buy from firms having no local physical presence.
- *Outsourcing is easier*: for similar reasons.

- *Digital marketing*: makes it easy and cost-effective to reach very specific groups of customers, regardless of location.
- *Network effects*: (discussed later) have become more powerful and accessible to firms.

Business Model Variations

There are many business model variations such as:

- *Private label or "contract" manufacturers* that produce goods to be marketed by others.
- *Licensing arrangements* in which a company will produce a product using someone else's brand name in return for a royalty. For example, toys and apparel based on famous film characters.
- *Value added resellers* that not only distribute a product but also handle more complex aspects of product installation, customization, service, or support. For example, heating/air conditioning systems resellers.
- *Franchise models* in which distributers or retailers have a tightly defined and often exclusive relationship with the parent company.

E-Commerce Business Models

E-commerce is a broad category that includes a wide range of internet-based direct sales models. A few key variations of e-commerce business models include:

- *Affiliate marketing* generates commission revenue from sales made on other people's websites. E.g., CJ Affiliate.
- *Marketplace businesses* create networks of buyers and sellers without taking ownership of the goods during the process. E.g., eBay.
- *Aggregators* are similar to marketplaces, but they re-markets products and services under their own brand. E.g., Uber.

Network Effects and Platform Business Models

The term "network effects" refers to the increase in value of a network to its users as more people join. Many internet-based businesses are built on network effects. For example, social media, ride-sharing services, online classified etc. Network effects are also applicable to older, non-internet businesses such as telephone services, credit cards etc.

Network effects can be described as:

- *Multi-sided*: applicable to two or more groups of users. For example, buyers and sellers on an online marketplace.
- *One-sided*: Users are a single, homogeneous group.

A **platform business** is defined as a business based on network effects—that is, where the value of its service or product is enhanced by the addition of customers or users. They are different from traditional or **linear** businesses. With a linear business, value is added by the firm; with a platform business, value is created in the network, outside the firm.

Crowdsourcing Business Models

Crowdsourcing business models enable users to contribute directly to a product, service, or online content. For example, Wikipedia, Google Maps, Tripadvisor.

Hybrid Business Models

Some companies employ hybrid models that combine platform and traditional "linear" businesses. For example, Tesla sells cars via a linear model, but its customers benefit from an expanding network of charging stations.

Summary

LO. Describe key features of business models.

A business model describes how a business is organized to deliver value to its customers:

- who its customers are,
- how the business serves them,
- key assets and suppliers, and
- the supporting business logic.

A business model should have a value proposition and a value chain.

- The firm's "value proposition" refers to the product or service attributes valued by a firm's target customer that lead those customers to prefer a firm's offering over those of its competitors, given relative pricing.
- The firm's "value chain" refers to how the firm is structured to deliver that value. It refers to the systems and processes within a firm that create value for its customers.

A firm's channel strategy refers to "where" the firm is selling its offering and how it is reaching its customers. Firms may employ a traditional channel strategy - using wholesalers and retailers or a direct sales strategy – selling directly to end customers. An omnichannel strategy refers to using both digital and physical channels to complete a sale.

Pricing is a key element of business models. Based on pricing power companies can be classified into price takers and price setters. Pricing approaches are typically value based or cost based.

To understand the profitability of a business, the analyst should examine margins, break-even points and unit economics (which is expressing revenues and costs on a per-unit basis).

LO: Describe various types of business models.

Each industry tends to has its own set of established business models. Firms in the goods-producing sectors are generally easy to classify based on how they fit into the supply chain. However, service businesses are more diverse.

Digital technology has enabled significant business model innovation, often based on network effects.

